



EEE 1101

Electrical Circuits

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Course Content

Basic Circuit Concept:

Basic electrical elements and quantities, Series and parallel circuits, Circuit Laws and Theorems, Mesh and Node analysis, Circuit with dependent sources, Capacitors and Inductors, Fundamental of AC Circuit, Phasor Algebra, AC circuit analysis, Series and parallel resonance, Influence of Circuit Parameters on Resonant Behavior, Q of a circuit, Wave trap, Series-Parallel Tuning.

Balanced Polyphase Circuits:

Two and four phase system, Three-phase four-wire system, Three-phase three-wire system, N-phase star and mesh, Balanced wye and delta loads, Three origin vector diagram of a balanced three phase system, Power measurement (both active and reactive) in balanced systems, Copper required to transmit power under fixed conditions.

Unbalanced Polyphase Circuits:

Unbalanced wye and delta loads, Network solutions, Phase-sequence effects, Methods of checking voltage phase sequence, Power measurements (both active and reactive) in unbalanced three phase systems



Reference Books

1. Fundamentals of Electric Circuits (5th edition)
 - Charles K. Alexander
 - Matthew N. O. Sadiku
2. Electric Circuits (10th edition)
 - James W. Nilsson
 - Susan A. Riedel
3. Alternating Current Circuits (4th edition)
 - Russel M. Kerchner
 - George F. Corcoran
4. Introductory Circuit Analysis (11th edition)
 - Robert L. Boylestad

Node and Mesh analysis



Example 10.1: Determine i_x using nodal analysis.

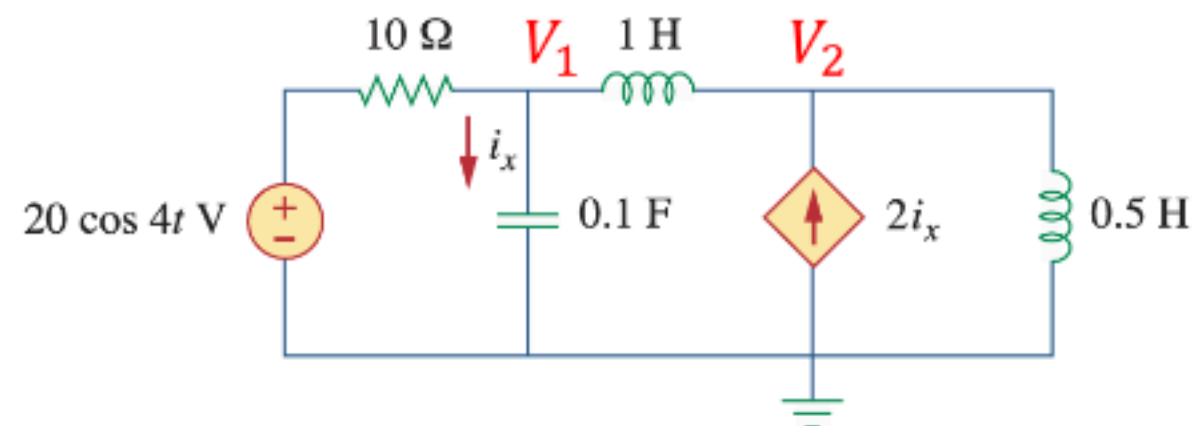


Figure 10.1
For Example 10.1.

$$(0.1 + j0.15) V_1 + j0.25 V_2 = 2 \quad \dots\dots\dots(i)$$

$$-j0.55 V_1 + (-j0.75) V_2 = 0 \quad \dots\dots\dots(ii)$$

$$AV_1 + BV_2 = 2 \quad \dots\dots\dots(i)$$

$$CV_1 + DV_2 = 0 \quad \dots\dots\dots(ii)$$

Solving equation (i) & (ii),

$$V_1 = \Delta_x / \Delta = 18.97 \angle 18.43^\circ \text{ V}$$

$$V_2 = \Delta_y / \Delta = 13.91 \angle 198.3^\circ \text{ V}$$

$$I_x = V_1 / (-j2.5) = 7.59 \angle 108.4^\circ \text{ A}$$

$$i_x = 7.59 \cos(4t + 108.4^\circ) \text{ A}$$

$$\Delta = \begin{vmatrix} A & B \\ C & D \end{vmatrix} = AD - BC = -0.25 - j0.075$$

$$\Delta_x = \begin{vmatrix} 2 & B \\ 0 & D \end{vmatrix} = 2D - B \cdot 0 = -j1.5$$

$$\Delta_y = \begin{vmatrix} A & 2 \\ C & 0 \end{vmatrix} = A \cdot 0 - 2C = j1.1$$

Node and Mesh analysis



10.39 Find $\mathbf{I}_1$, $\mathbf{I}_2$, $\mathbf{I}_3$, and $\mathbf{I}_x$ in the circuit of Fig. 10.84.

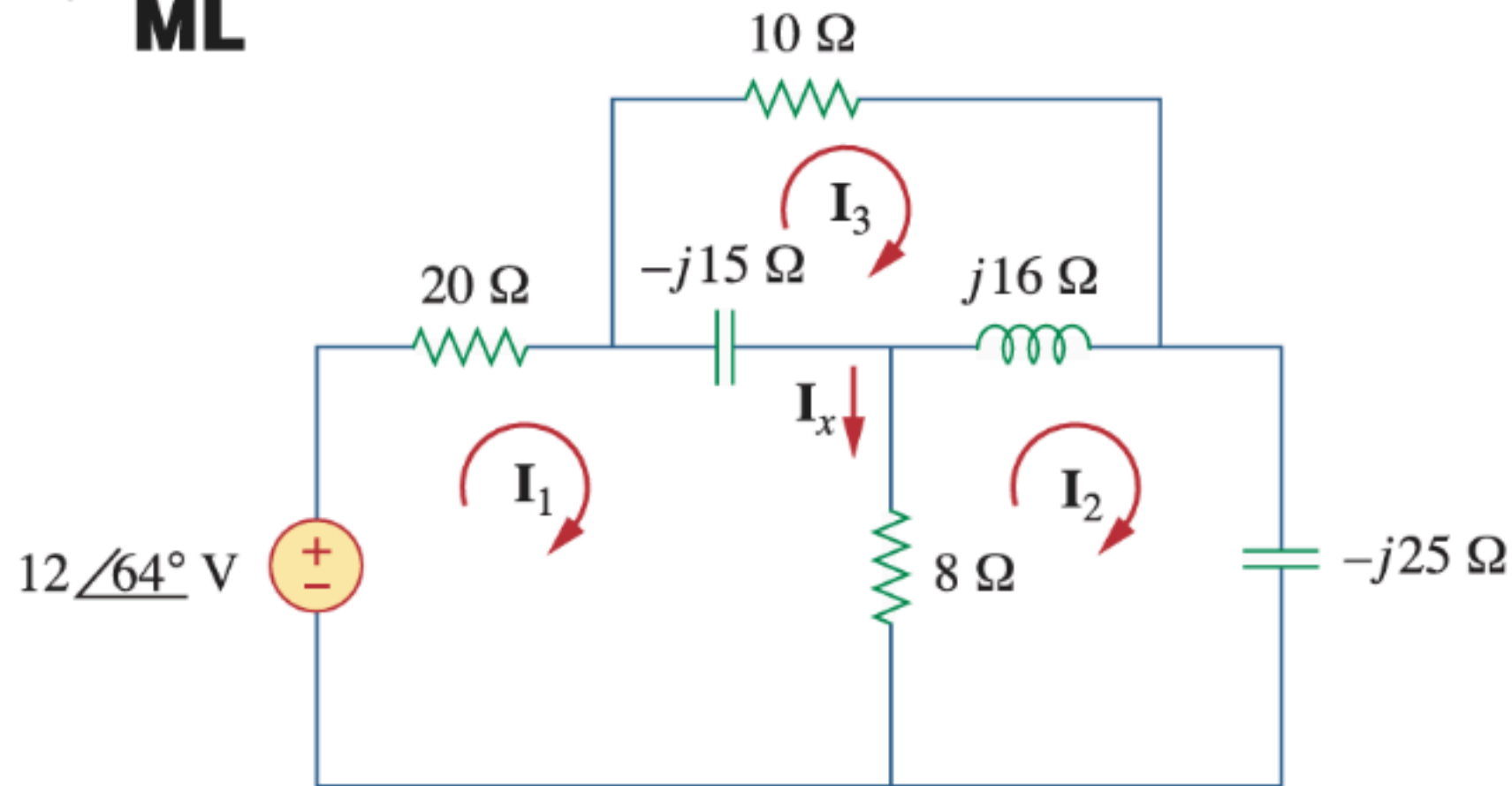


Figure 10.84
For Prob. 10.39.

$$(28-j15) I_1 + (-8) I_2 + j15 I_3 = 12\angle 64^\circ \quad \dots(i)$$

$$(-8) I_1 + (8-j9) I_2 + (-j16) I_3 = 0 \quad \dots(ii)$$

$$j15 I_1 + (-j16) I_2 + (10+j) I_3 = 0 \quad \dots(iii)$$

$$A I_1 + B I_2 + C I_3 = 12\angle 64^\circ \quad \dots(i)$$

$$B I_1 + D I_2 + E I_3 = 0 \quad \dots(ii)$$

$$C I_1 + E I_2 + F I_3 = 0 \quad \dots(iii)$$



$$\begin{aligned} A I_1 + B I_2 + C I_3 &= 12\angle 64^\circ \dots(i) \\ B I_1 + D I_2 + E I_3 &= 0 \dots\dots\dots(ii) \\ C I_1 + E I_2 + F I_3 &= 0 \dots\dots\dots(iii) \end{aligned}$$

$$\Delta = \begin{vmatrix} A & B & C \\ B & D & E \\ C & E & F \end{vmatrix} = A(DF-EE) - B(BF-CE) + C(BE-CD) = 5750 - j9560$$

$$\Delta_x = \begin{vmatrix} 12\angle 64^\circ & B & C \\ 0 & D & E \\ 0 & E & F \end{vmatrix} = 12\angle 64^\circ (DF-EE) - B(0.F-0.E) + C(0.E-0.D) = 2699.27 + j3289.65$$

$$\Delta_y = \begin{vmatrix} A & 12\angle 64^\circ & C \\ B & 0 & E \\ C & 0 & F \end{vmatrix} = A(0.F-0.E) - 12\angle 64^\circ (BF-CE) + C(B.0-C.0) = 1597.06 + j3493.45$$

$$\Delta_z = \begin{vmatrix} A & B & 12\angle 64^\circ \\ B & D & 0 \\ C & E & 0 \end{vmatrix} = A(D.0-E.0) - B(B.0-C.0) + 12\angle 64^\circ (BE-CD) = -796.44 - j1413.96$$

$$\begin{aligned} I_1 &= \Delta_x / \Delta = 0.3814\angle 109.6^\circ \text{ A} \\ I_2 &= \Delta_y / \Delta = 0.3443\angle 124.4^\circ \text{ A} \\ I_3 &= \Delta_z / \Delta = 0.1454\angle -60.41^\circ \text{ A} \\ I_x &= I_1 - I_2 = 0.1004\angle 48.48^\circ \text{ A} \end{aligned}$$

Node and Mesh analysis



Homework:

From: *Fundamentals of Electric Circuits* (5th edition)

- Example: 10.1–10.4
- practice problem: 10.1–10.4
- Exercise: 10.8, 10.9, 10.10, 10.15, 10.18, 10.37

From: *Electric Circuits* (10th edition)

- Example: 9.11, 9.12

Assignment

10.19 Obtain V_o in Fig. 10.68 using nodal analysis.

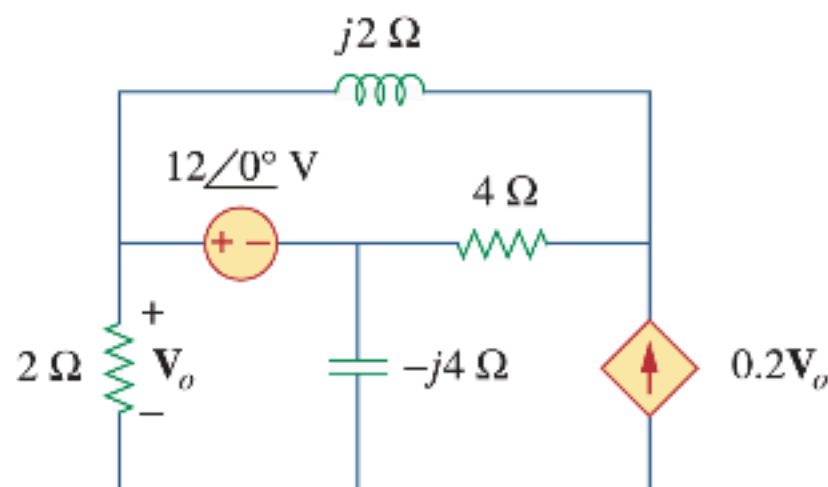


Figure 10.68

For Prob. 10.19.

10.38 Using mesh analysis, obtain I_o in the circuit shown in Fig. 10.83.

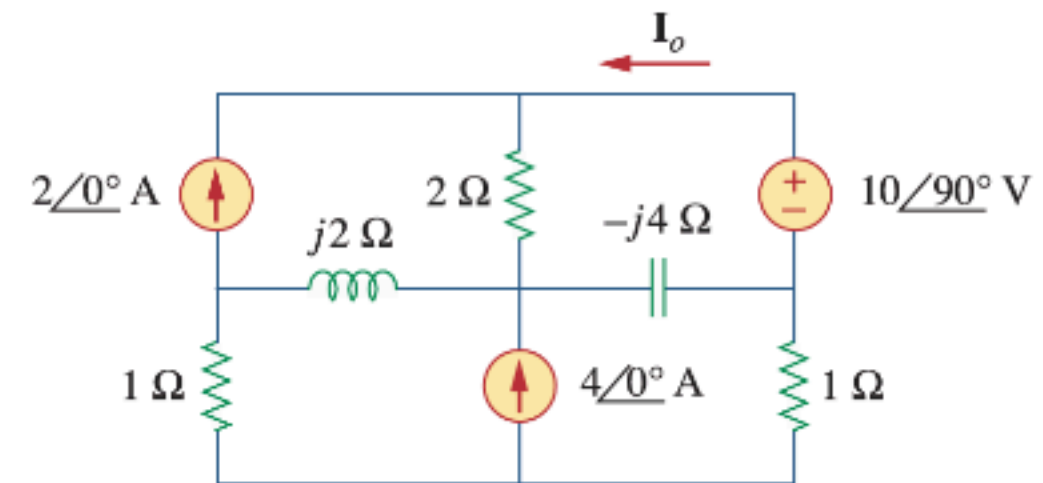


Figure 10.83

For Prob. 10.38.

Superposition Theorem



Example 10.6: Find v_o of the circuit of Fig. 10.13 using the superposition theorem.

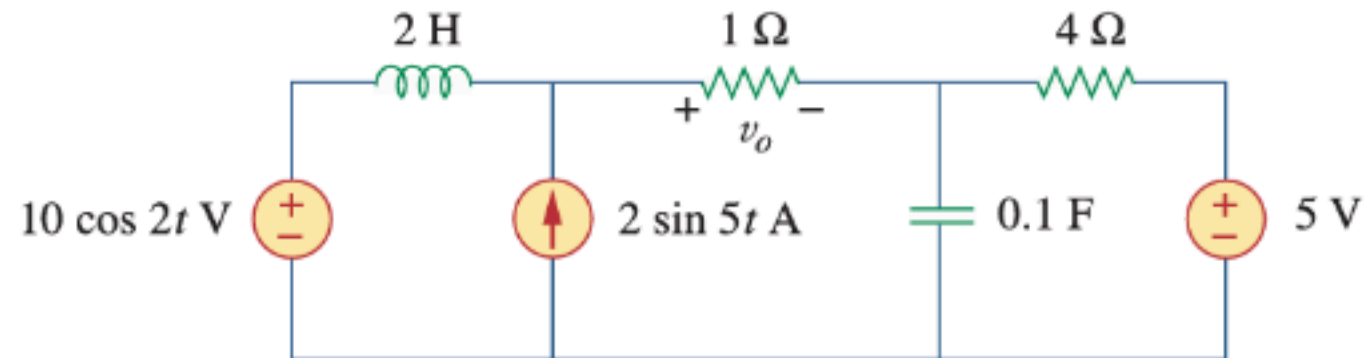
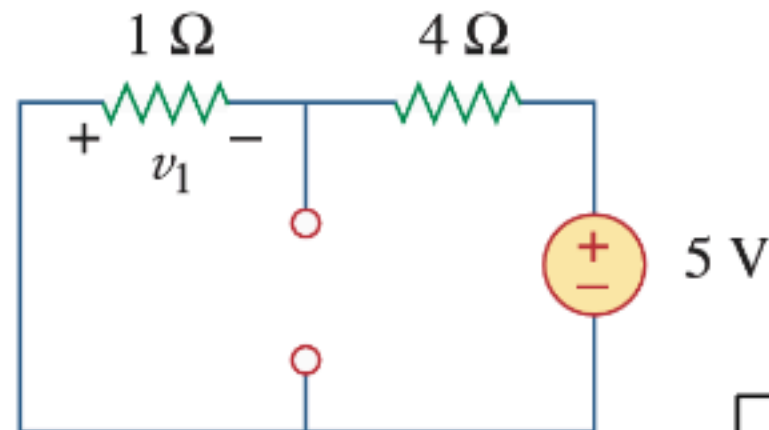


Figure 10.13
For Example 10.6.

Solution:

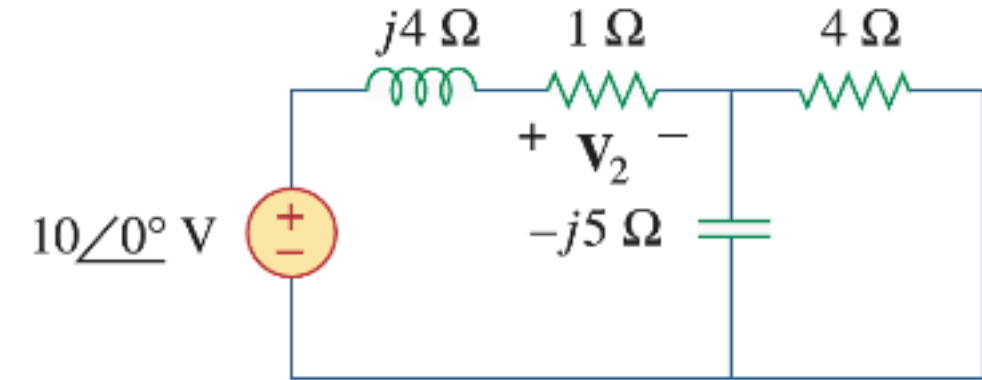
5V source Active:



(a)

$$V_1 = -1V$$

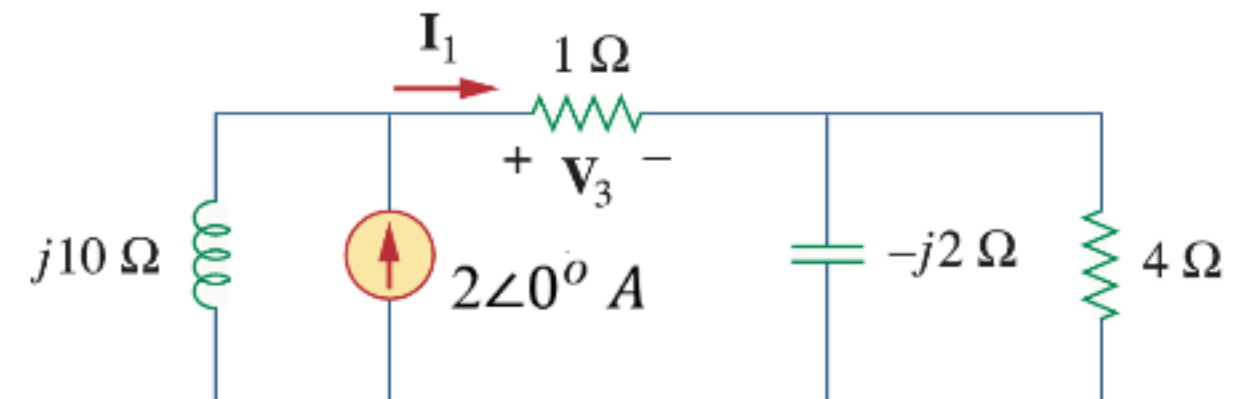
10cos2t V source Active:



(b)

$$V_2 = 2.498\angle -30.79^\circ \text{ and } v_2 = 2.498\cos(2t - 30.79^\circ) \text{ V}$$

10sin5t V source Active:



(c)

$$V_2 = 2.33\angle 12^\circ \text{ and } v_2 = 2.33\sin(5t + 12^\circ) \text{ V}$$

$$v_o = v_1 + v_2 + v_3 = -1 + 2.498\cos(2t - 30.79^\circ) + 2.33\sin(5t + 12^\circ) \text{ V}$$



Thevenin and Norton Theorem

Example 10.9: Find the Thevenin equivalent of the circuit in Fig. 10.25 as seen from terminals a-b.

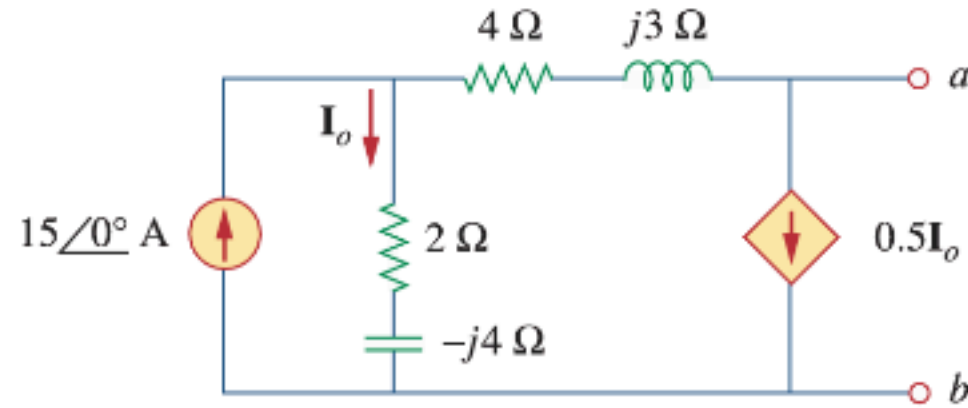
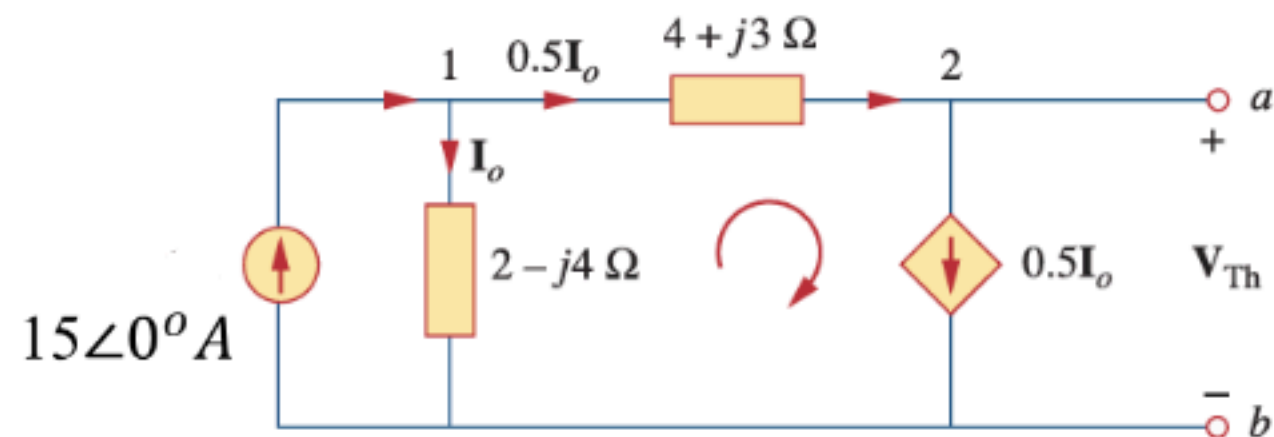


Figure 10.25
For Example 10.9.

Solution:

Calculation of Thevenin voltage (V_{th}):



(a)

Applying KCL at node 1,
 $15\angle 0^\circ = I_o + 0.5 I_o$
 $\Rightarrow I_o = 10\angle 0^\circ A$

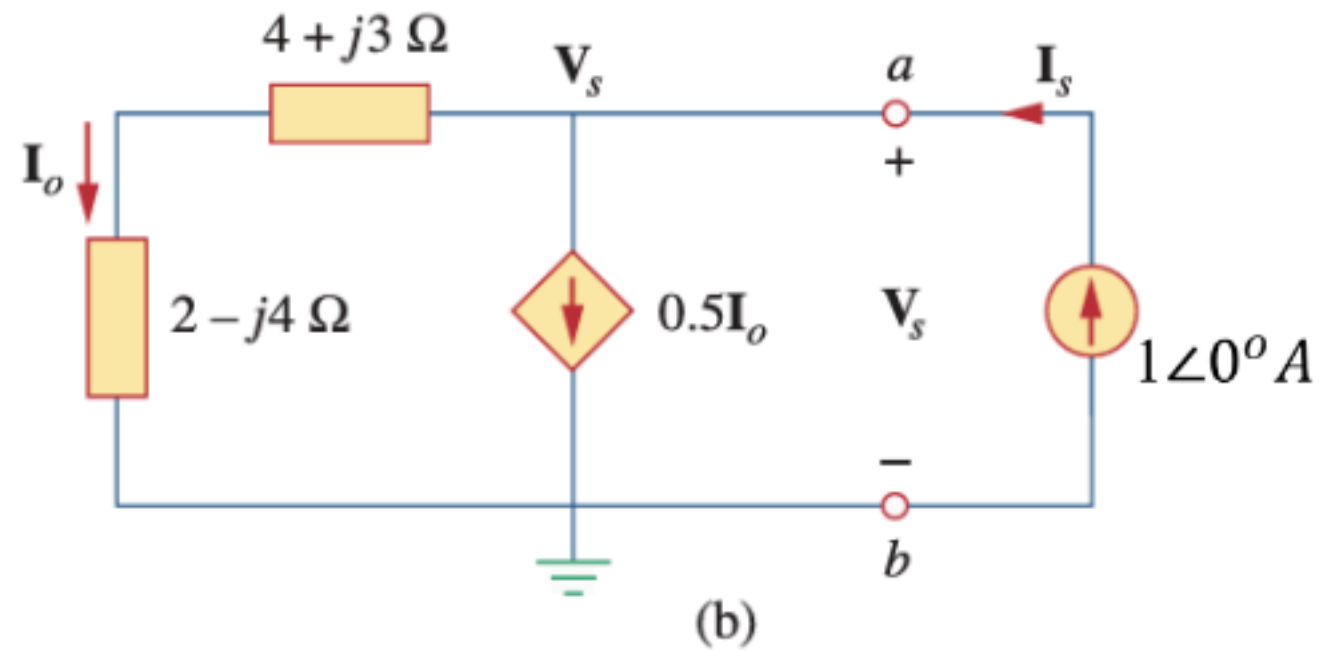
Applying KVL,
 $-V_{th} - (4+j3) 0.5 I_o + (2-j4) I_o$
 $\Rightarrow V_{th} = 55\angle -90^\circ V$

Note: When one or more independent sources are present in the circuit, we cannot turn it off because it depends on other circuit variables. It must stay active when finding the Thevenin impedance (z_{th}).



Thevenin and Norton Theorem

Calculation of Thevenin impedance (Z_{th}):



Applying KVL,

$$-V_s + (4+j3)I_o + (2-j4)I_o$$

$$\Rightarrow V_s = (4-j\frac{2}{3})V$$

Applying KCL,

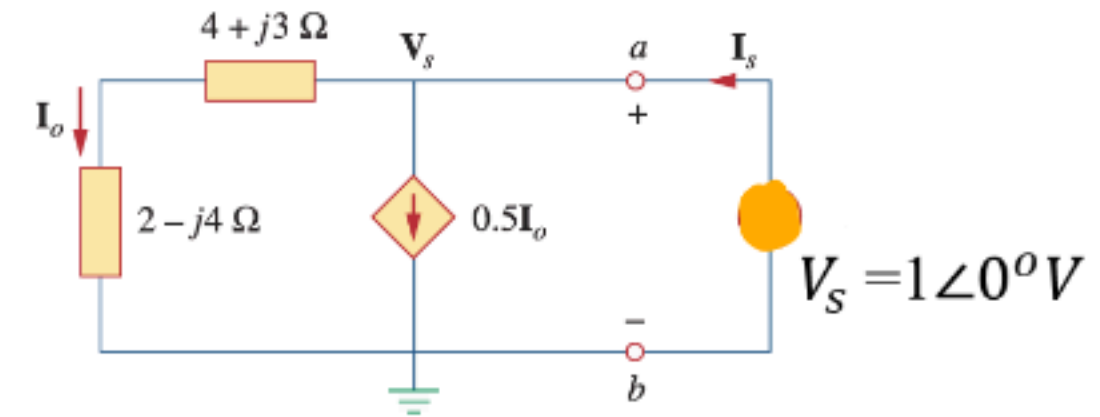
$$1\angle 0^\circ = I_o + 0.5 I_o$$

$$\Rightarrow I_o = \frac{2}{3}\angle 0^\circ A$$

Thevenin Impedance,

$$Z_{th} = V_s / I_s = (4-j\frac{2}{3})\Omega$$

Alternative way of finding Z_{th} :

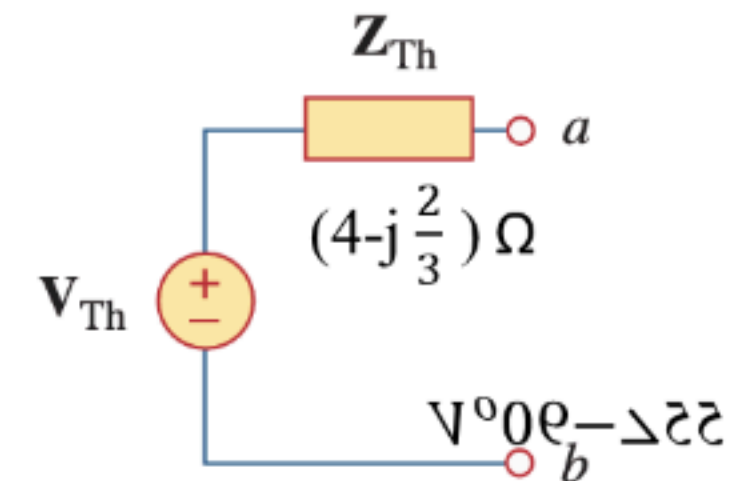


$$I_o = \frac{1\angle 0^\circ}{(4+j3) + (2-j4)} = (\frac{6}{37} + j\frac{1}{37})A$$

$$I_s = I_o + 0.5 I_o = (\frac{9}{37} + j\frac{3}{37})A$$

Thevenin Impedance,

$$Z_{th} = V_s / I_s = (4-j\frac{2}{3})\Omega$$



Thevenin and Norton Theorem

Find Norton equivalent circuit of example 10.9

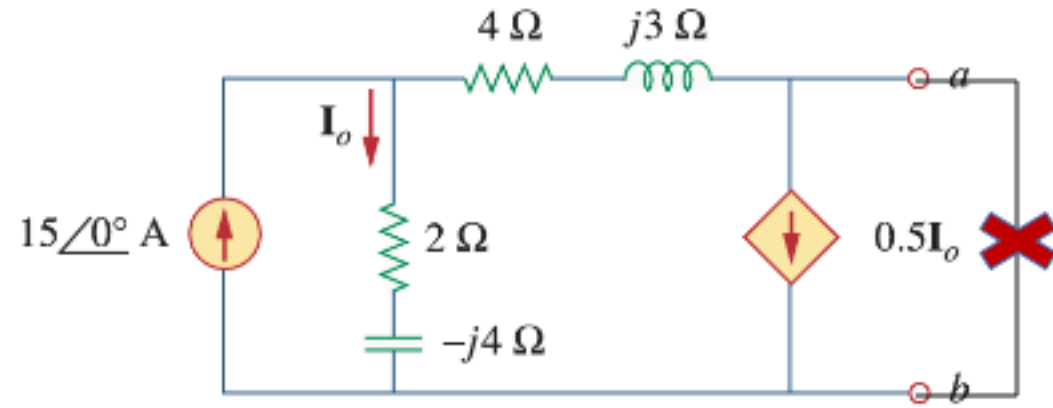
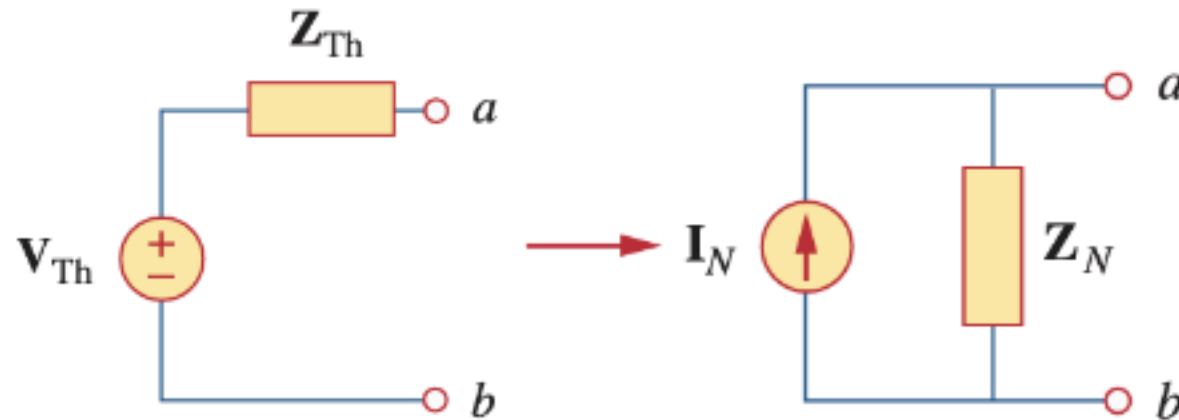


Figure 10.25
For Example 10.9.

The variable on which the dependent source depends can be made zero; but the dependent source itself cannot be made zero, because its value depends on another variable.

Solution:



Norton Impedance (Z_N) = Thevenin Impedance (Z_{th})

$$Z_N = (4 - j\frac{2}{3}) \Omega$$

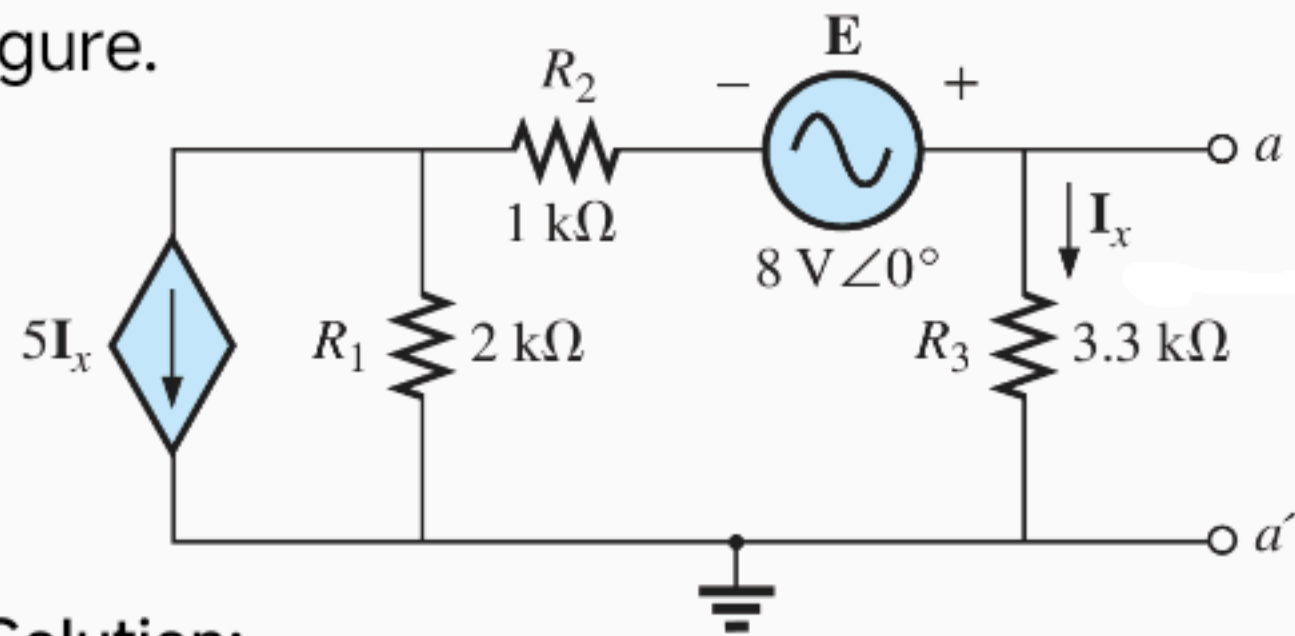
$$I_N = V_{th} / Z_{th} = (55 \angle -90^\circ) / (4 - j\frac{2}{3})$$

$$\Rightarrow I_N = 13.56 \angle -80.53^\circ \text{ A}$$



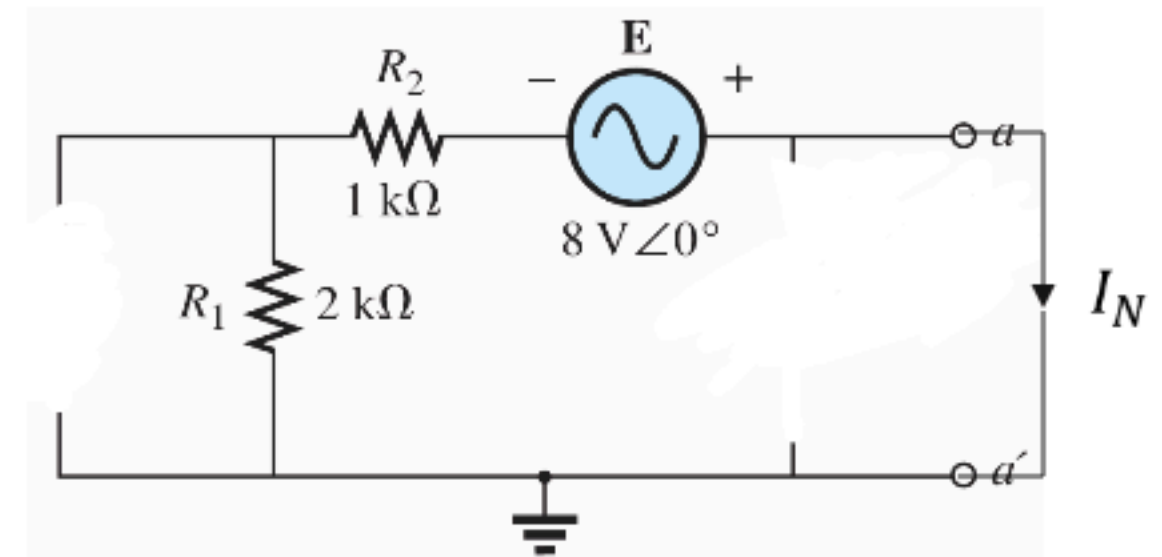
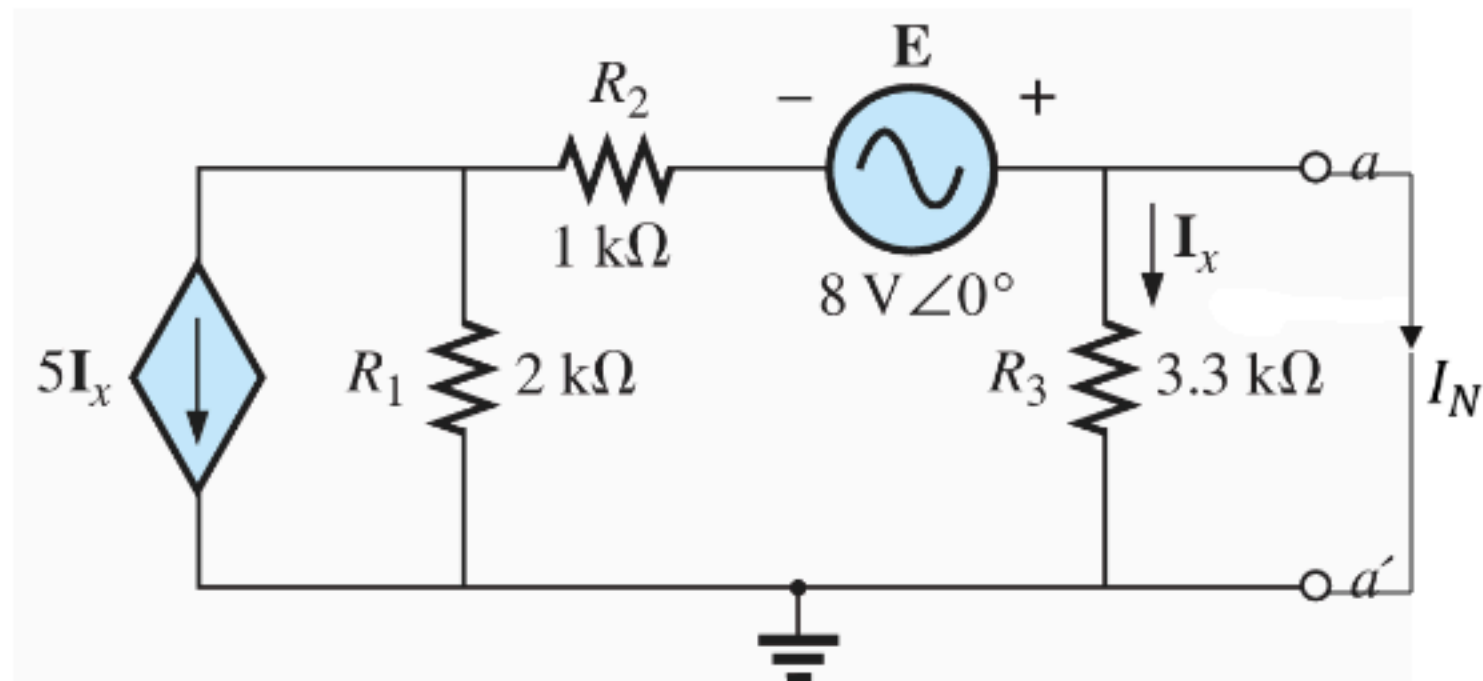
Thevenin and Norton Theorem

Find Norton equivalent circuit for the network to the left of terminals a-a' in Figure.



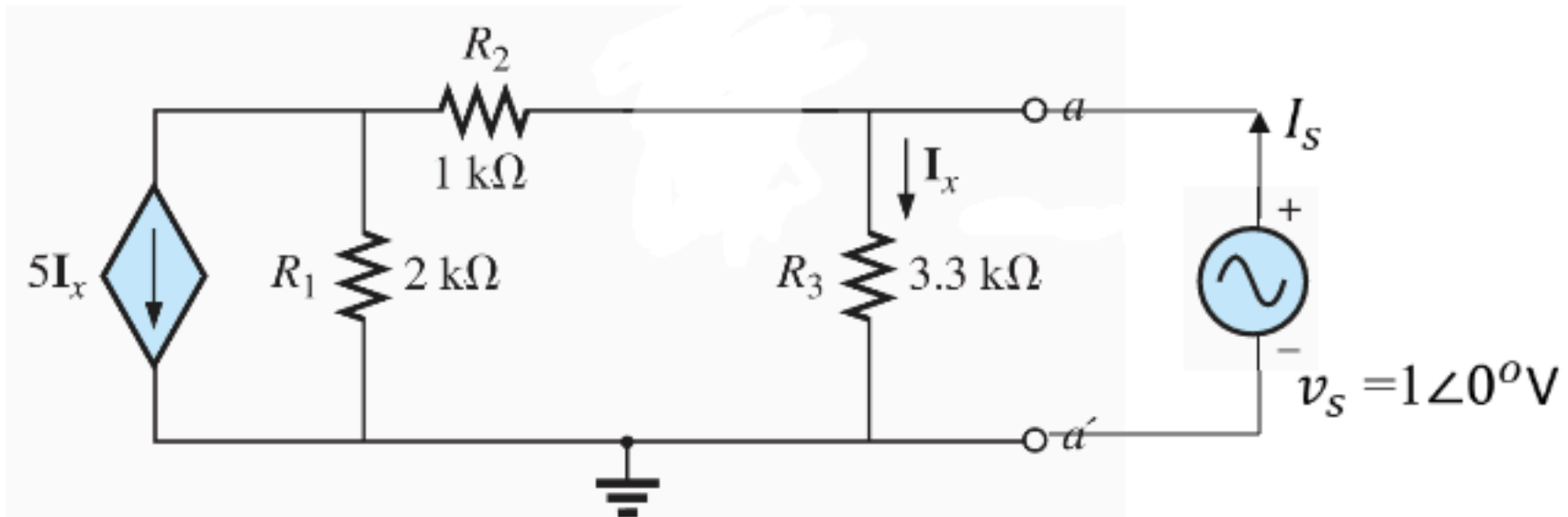
Solution:

Calculation of Norton Current (I_N):

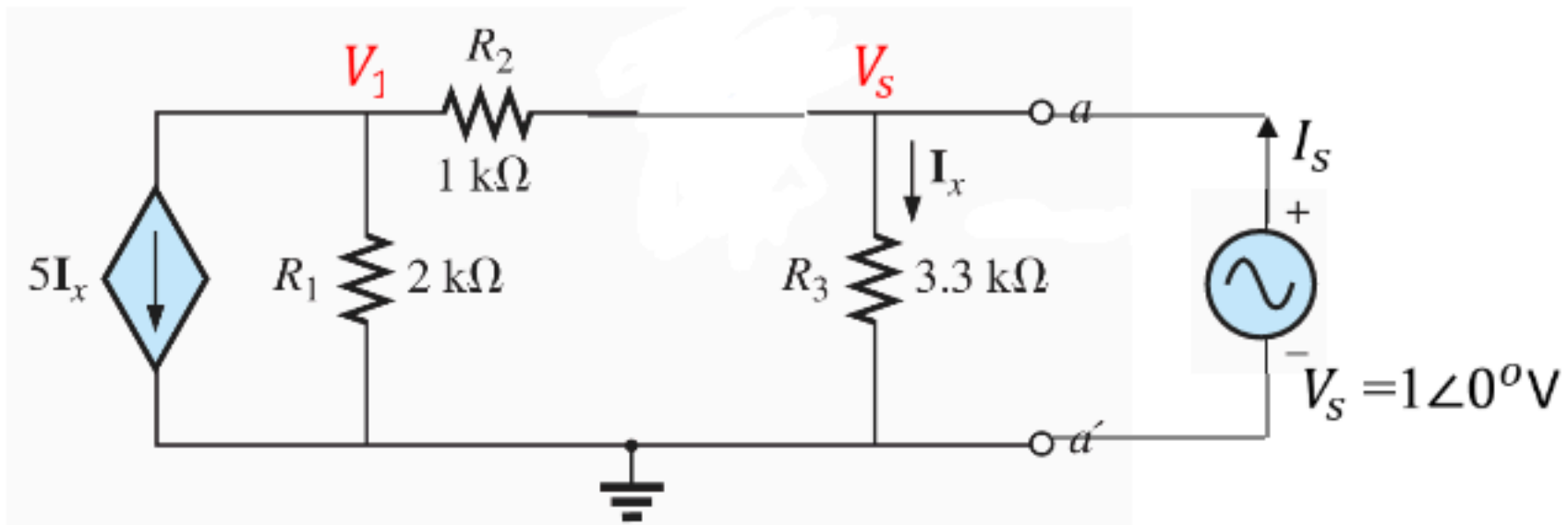


$$I_N = \frac{8 \angle 0^\circ}{(1+2)k} = \frac{8}{3} \angle 0^\circ \text{ mA} = 2.67 \angle 0^\circ \text{ mA}$$

Calculation of Norton impedance (Z_N):



Thevenin and Norton Theorem



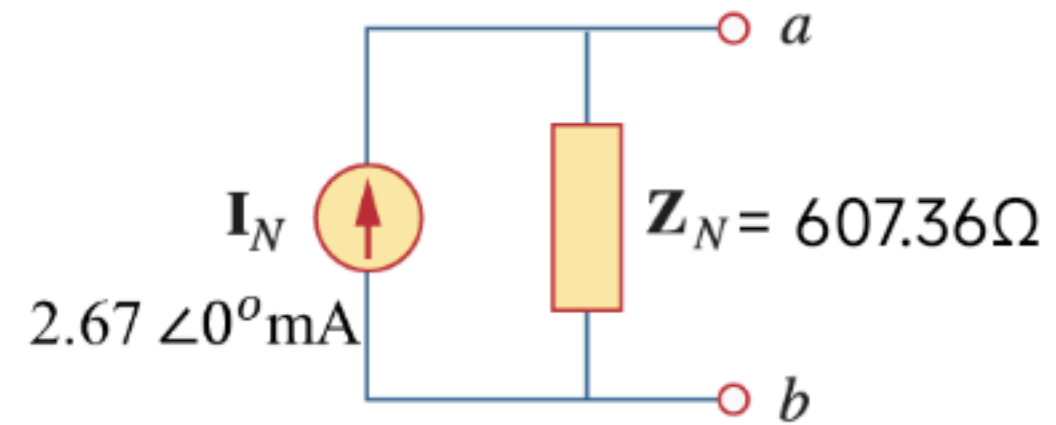
$$I_x = (1\angle 0^\circ) / (3.3\text{k}) = \frac{1}{3.3}\text{mA}$$

$$(V_1 - V_S)/1\text{k} + V_1/2\text{k} + 5I_x = 0$$

$$\Rightarrow V_1 = \frac{-34}{99}\text{V}$$

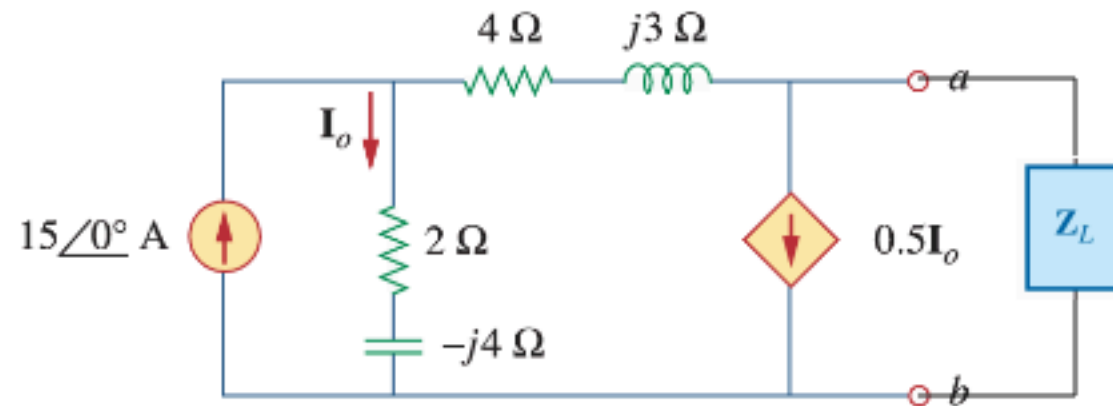
$$I_S = (V_S - V_1)/1\text{k} + I_x = 1.64\text{mA}$$

Norton Impedance,
 $Z_N = V_S / I_S = 607.36\Omega$

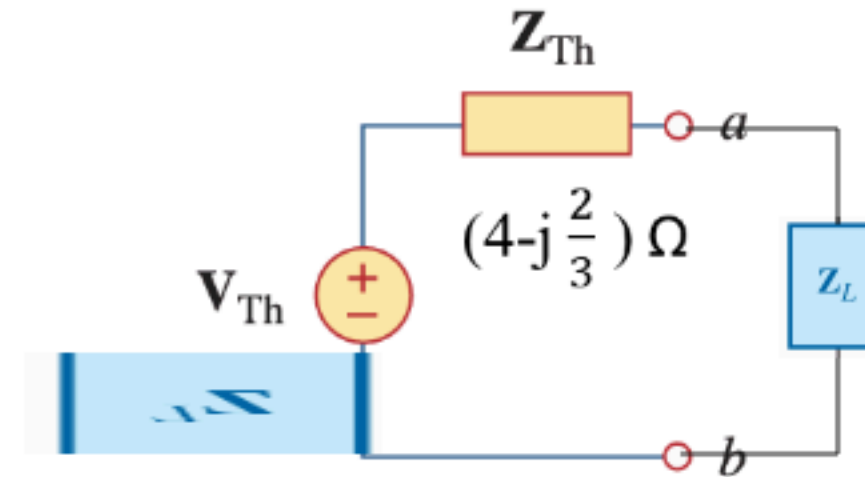
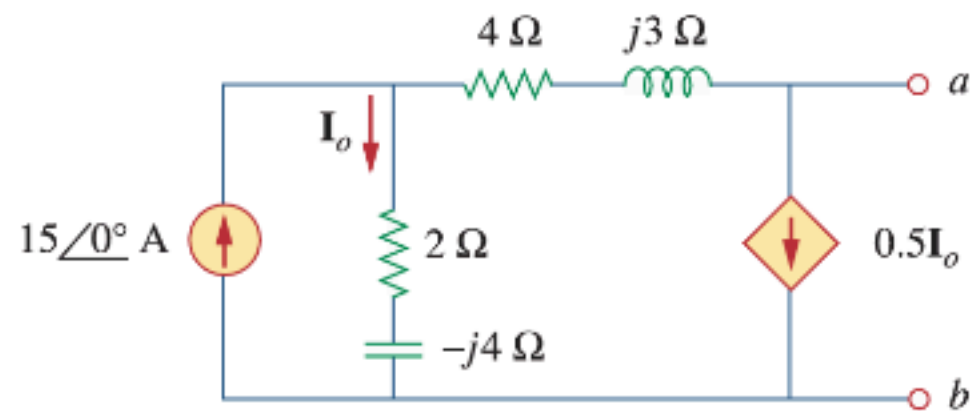


Maximum Power Transfer Theorem

Find the load impedance Z_L for the networks in Figure for maximum power to the load, and find the maximum power to the load.



Solution:



$$Z_L = (Z_{th})^* = (4 + j \frac{2}{3}) \Omega$$

If V_{th} is a rms value,

The maximum power of load, $P_m = (V_{th})^2 / (4R_{th})$

$$\Rightarrow P_m = 55^2 / (4 \times 4) = 189.06 \text{ W}$$

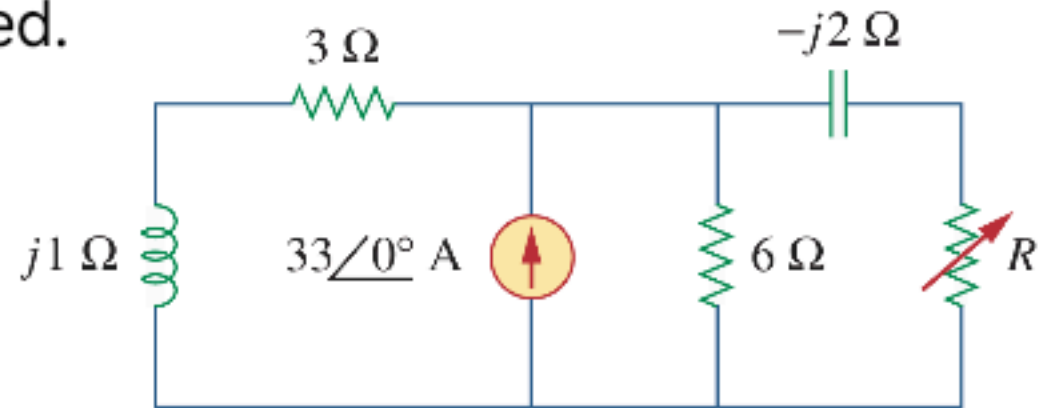
If V_{th} is a maximum value,

The maximum power of load, $P_m = (V_{th})^2 / (8R_{th})$

$$\Rightarrow P_m = 55^2 / (8 \times 4) = 94.53 \text{ W}$$

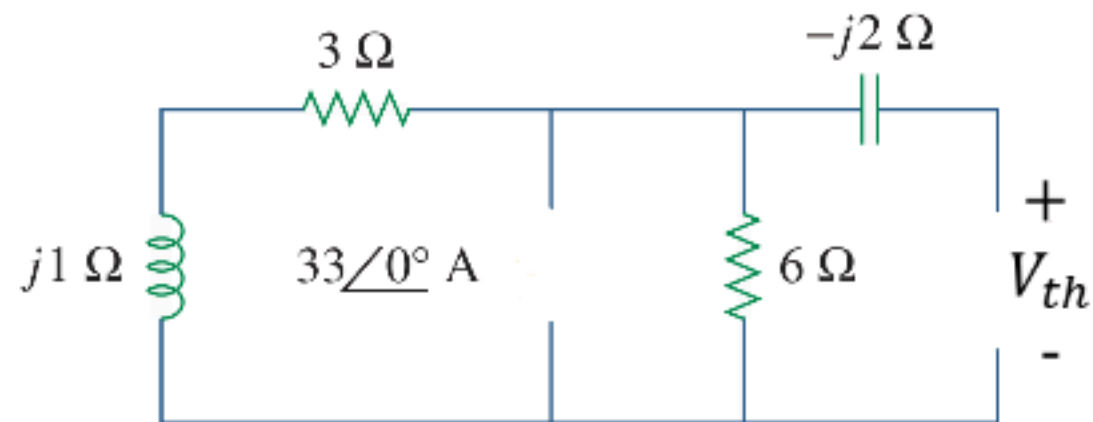
Maximum Power Transfer Theorem

The variable resistor R in the circuit of Fig. 11.50 is adjusted until it absorbs the maximum average power. Find R and the maximum average power absorbed.



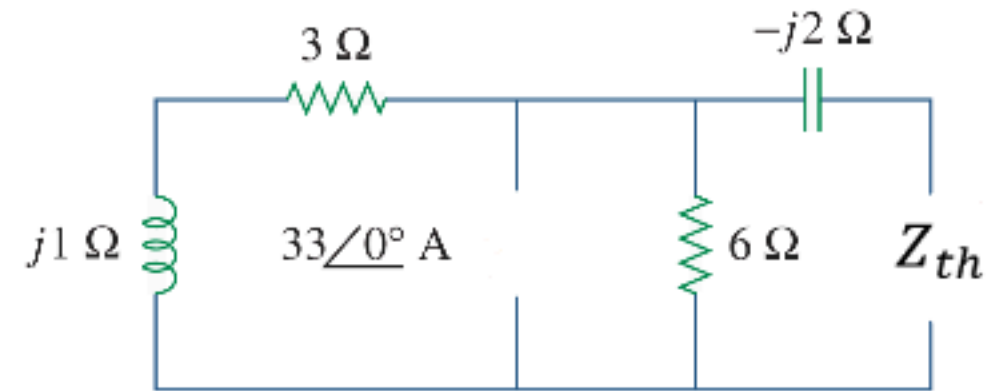
Solution:

Calculation of Thevenin voltage (V_{th}):



$$V_{th} = (33 \angle 0^\circ)((3+j1) \parallel 6) = 69.14 \angle 12.09^\circ \text{ V}$$

Calculation of Thevenin impedance (Z_{th}):



$$Z_{th} = -j2 + ((3+j1) \parallel 6) = (2.048-j1.56) \Omega = 2.575 \angle -37.3^\circ \Omega$$



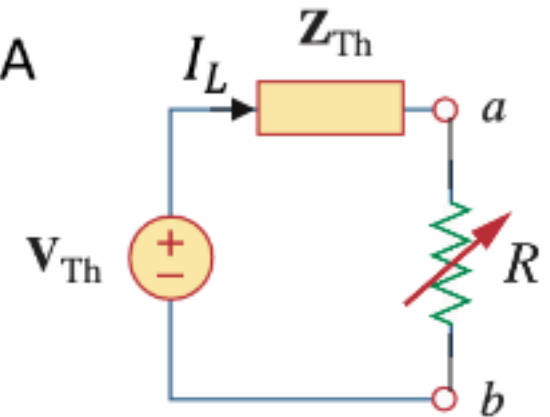
$$I_L = V_{th} / (Z_{th} + R) = 14.17 \angle 30.74^\circ \text{ A}$$

If V_{th} is a rms value,

The maximum power of load,

$$P_m = (I_L)^2 R_{th}$$

$$\Rightarrow P_m = 14.17^2 \times 2.575 = 517.03 \text{ W}$$



If V_{th} is a maximum value,

The maximum power of load,

$$P_m = (I_L)^2 R_{th} / 2 = 14.17^2 \times 2.575 / 2 = 258.51 \text{ W}$$

Assignment



10.68 Find the Thevenin equivalent at terminals $a-b$ in the circuit of Fig. 10.111.



ML

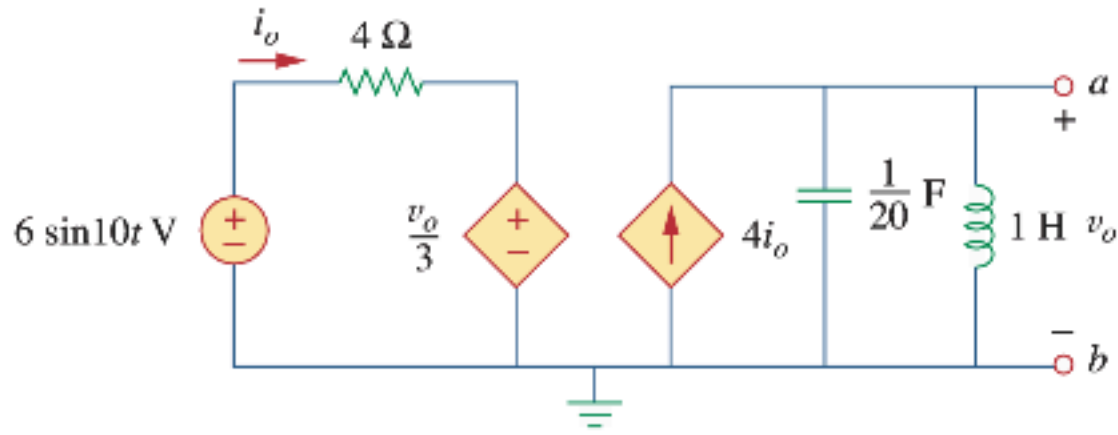


Figure 10.111

For Prob. 10.68.

11.16 For the circuit in Fig. 11.47, find the value of $\mathbf{Z}_L$ that will receive the maximum power from the circuit. Then calculate the power delivered to the load $\mathbf{Z}_L$.

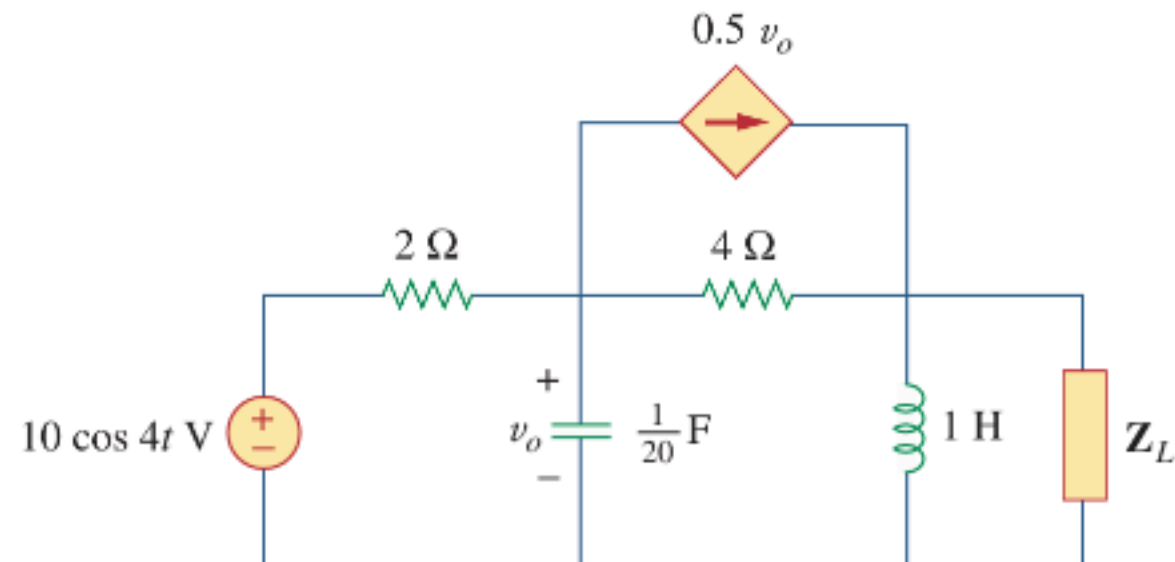


Figure 11.47

For Prob. 11.16.

10.68 Find the Thevenin equivalent at terminals $a-b$ in the circuit of Fig. 10.111.



ML

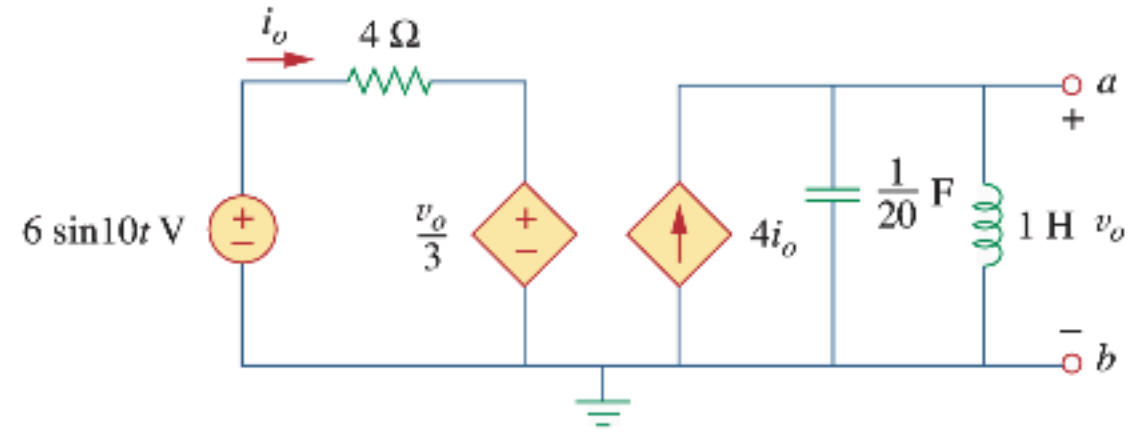


Figure 10.111

For Prob. 10.68.

11.20 The load resistance R_L in Fig. 11.51 is adjusted until it absorbs the maximum average power. Calculate the value of R_L and the maximum average power.

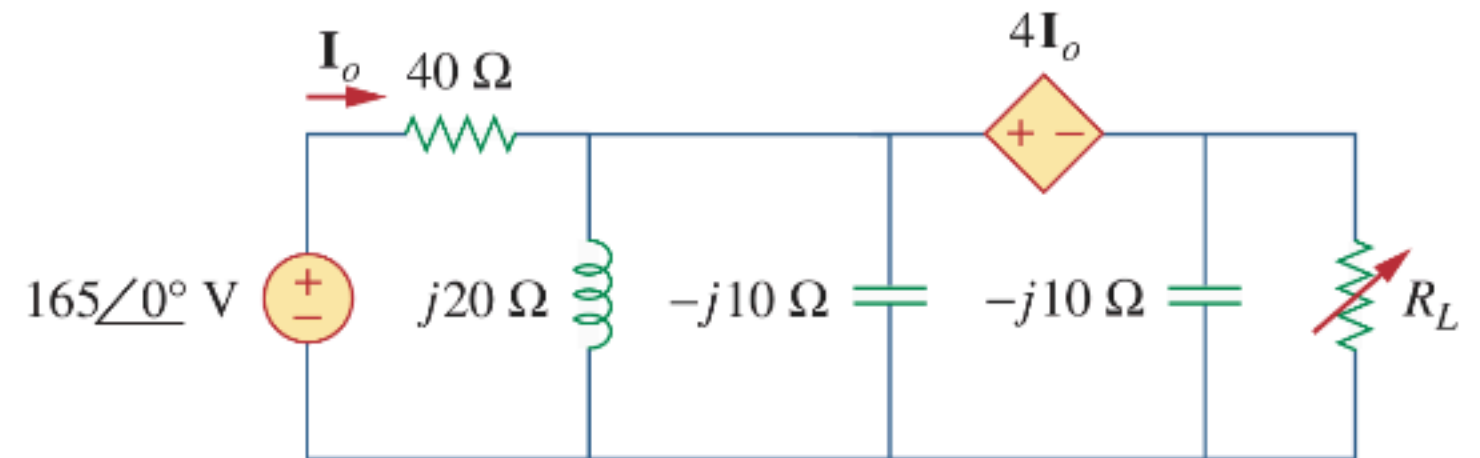


Figure 11.51

For Prob. 11.20.



Homework:

From: *Fundamentals of Electric Circuits (5th edition)*

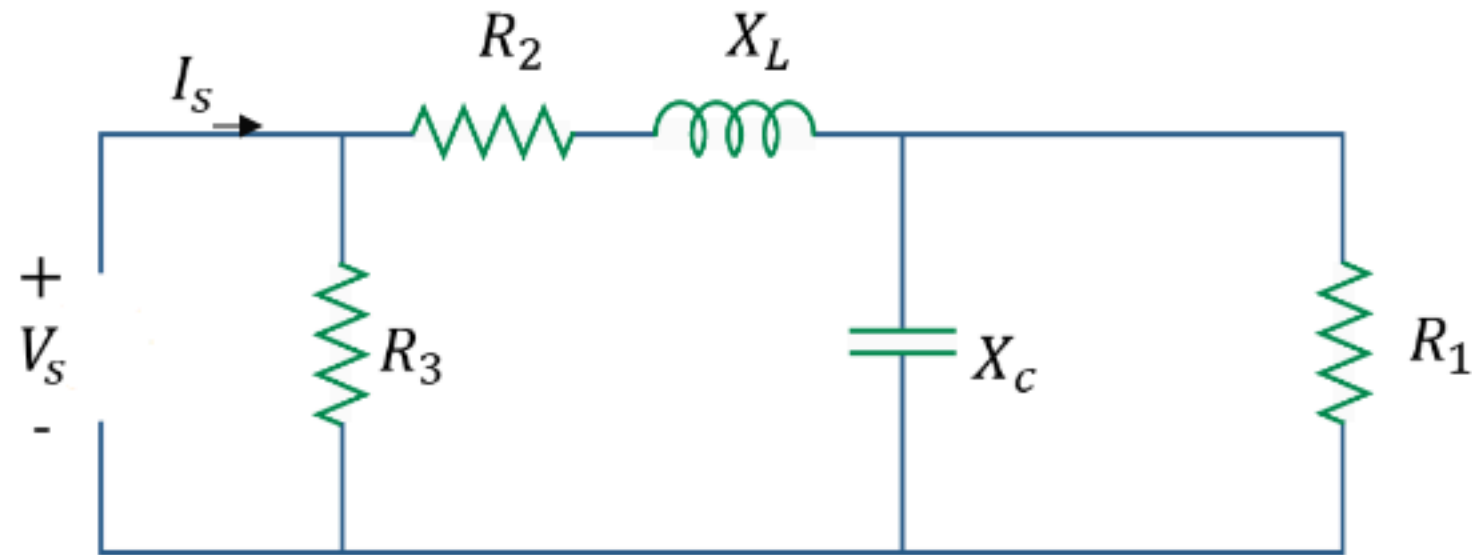
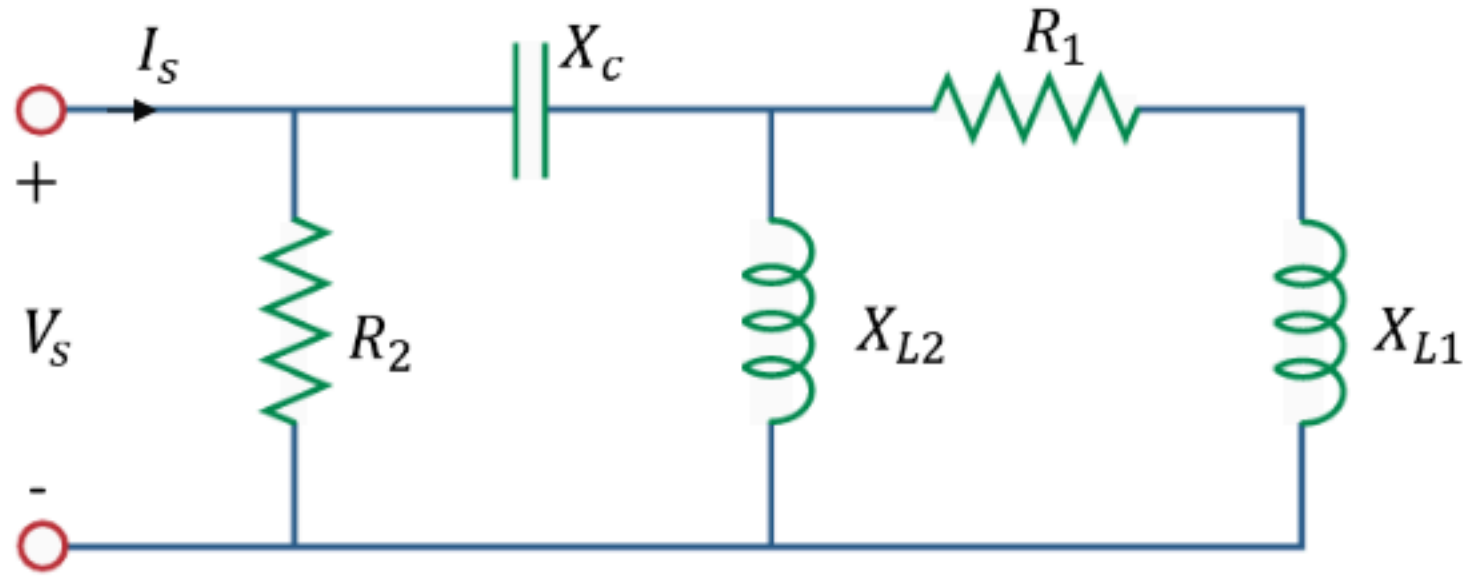
- Example: 10.9, 10.10, 11.5, 11.6
- practice problem: 10.9, 10.10, 11.5, 11.6
- Exercise: 10.47, 10.48, 10.61, 10.62, 11.15

Thank you.



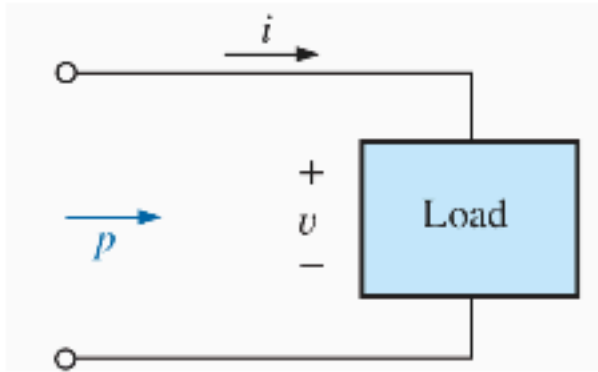
Phasor Diagram

Draw the phasor diagram of the given circuits:



Instantaneous Power

RESISTIVE
CIRCUIT:

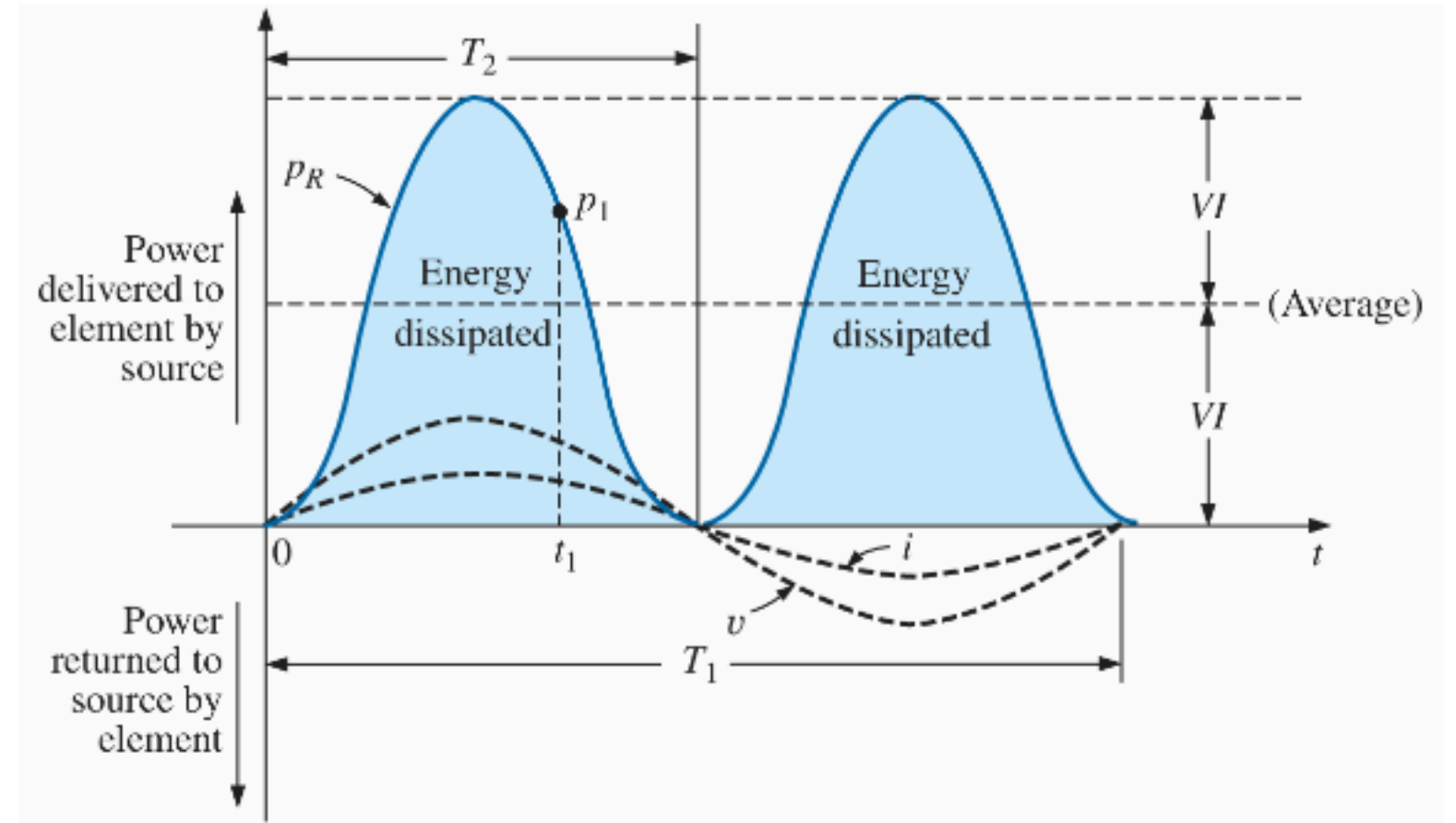


Instantaneous voltage, $v = V_m \sin \omega t$

Instantaneous current, $i = I_m \sin \omega t$

Instantaneous power,

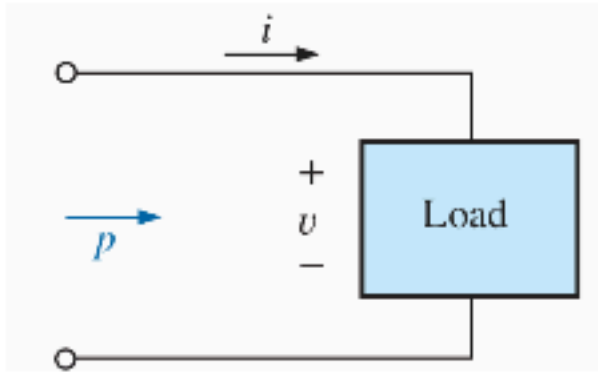
$$\begin{aligned} p &= vi = V_m \sin \omega t \cdot I_m \sin \omega t \\ &= \frac{1}{2} V_m I_m - \frac{1}{2} V_m I_m \cos 2 \omega t \\ &= VI - VI \cos 2 \omega t \end{aligned}$$



Power versus time for a purely resistive load.

Instantaneous Power

PURELY INDUCTIVE CIRCUIT:



Instantaneous current, $i = I_m \sin \omega t$

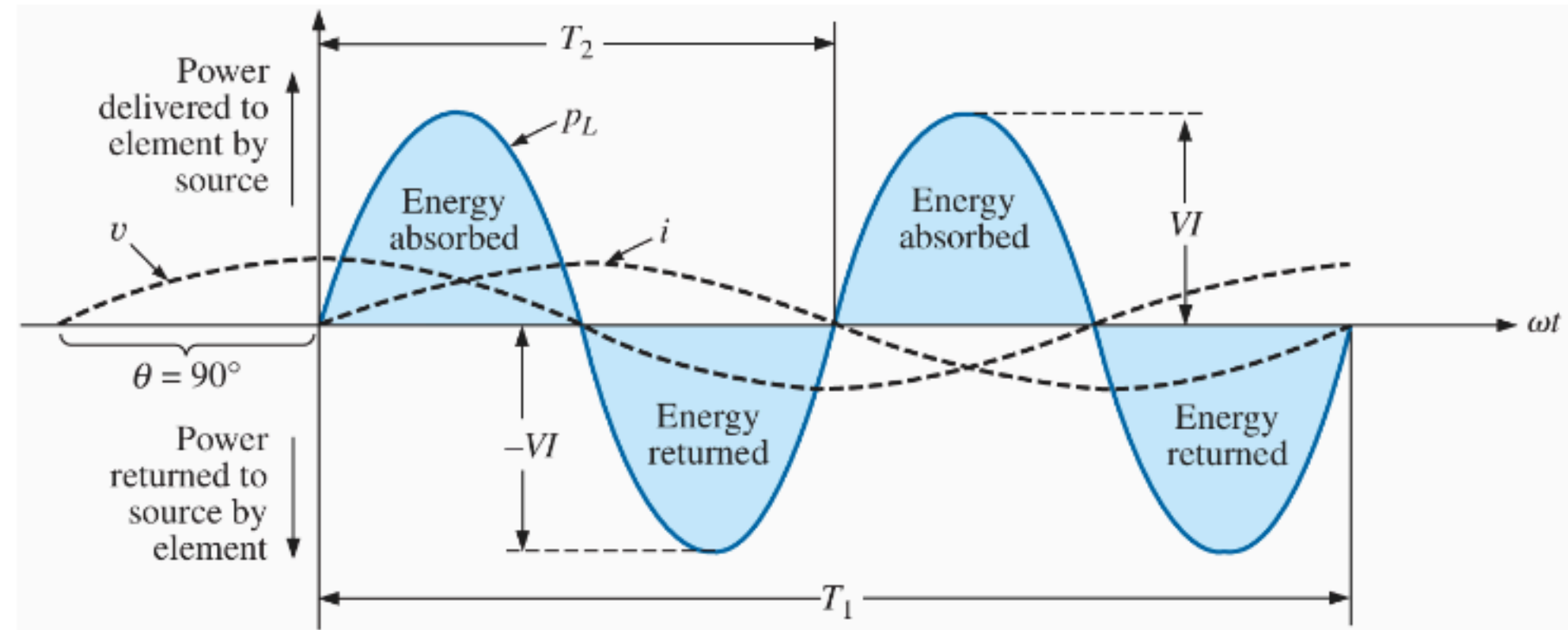
Instantaneous voltage, $v = V_m \sin(\omega t + 90^\circ)$

Instantaneous power,

$$p = vi = V_m \sin(\omega t + 90^\circ) \cdot I_m \sin \omega t$$

$$= \frac{1}{2} V_m I_m \sin 2 \omega t$$

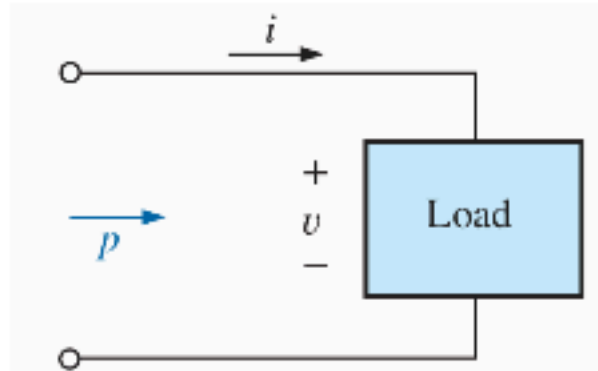
$$= V \sin 2 \omega t$$



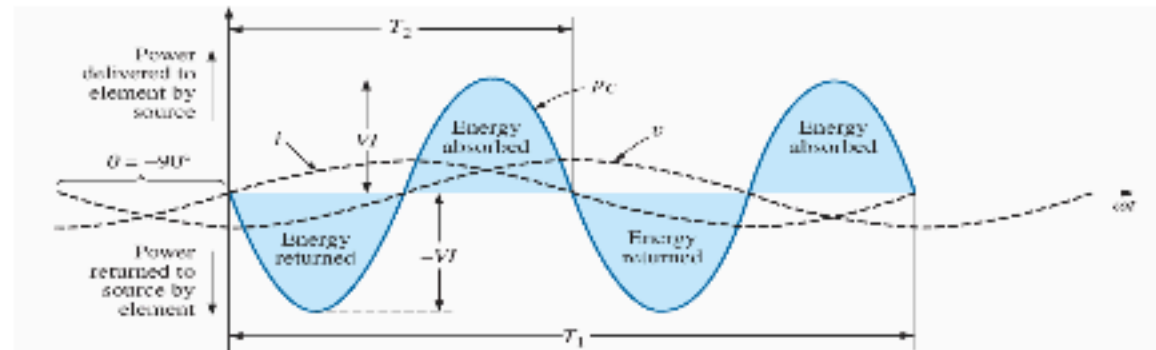
The power curve for a purely inductive load.

Instantaneous Power

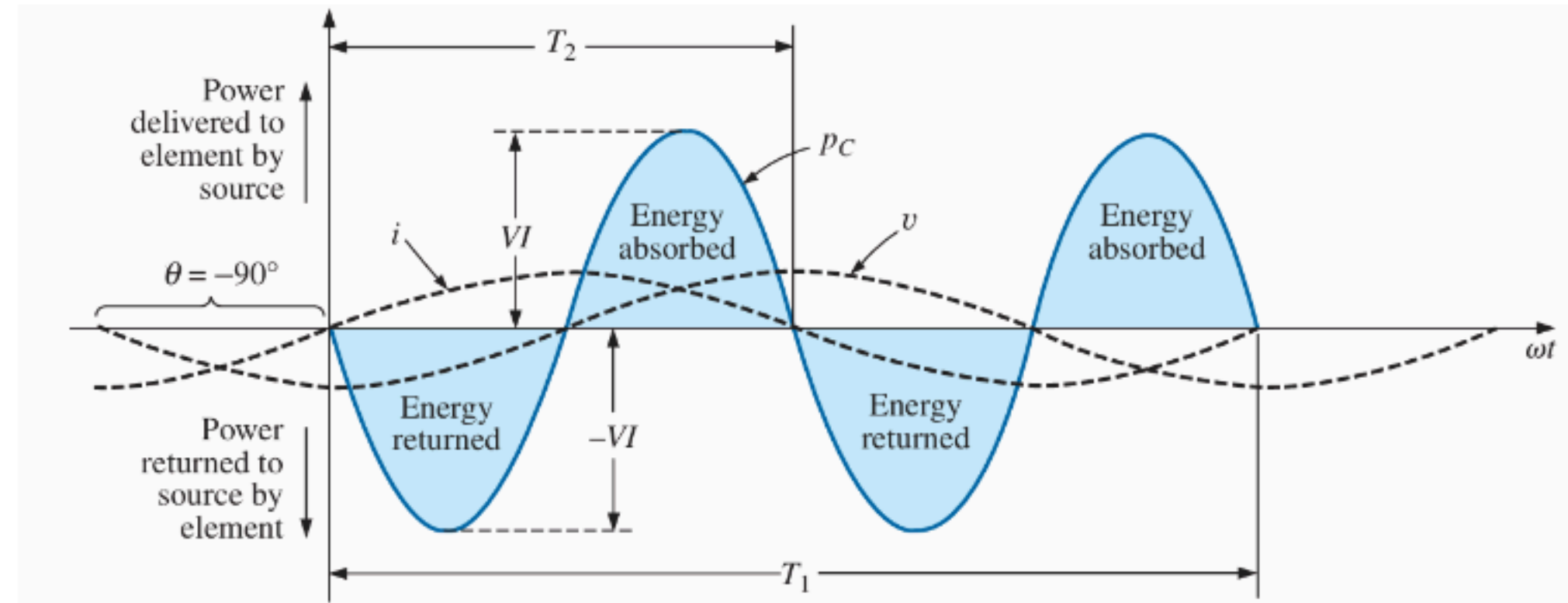
PURELY CAPACITIVE CIRCUIT:



Instantaneous current, $i = I_m \sin \omega t$
 Instantaneous voltage, $v = V_m \sin(\omega t - 90^\circ)$



The power curve for a purely capacitive load.



The power curve for a purely capacitive load.

Instantaneous Power

Question: For the series RL circuit shown in Fig. 1 with an applied sinusoidal voltage $v_s = 141.4 \sin \omega t$ volts, derive the expression for the instantaneous current (i_s) and the instantaneous power (p_s) consumed by the circuit.

Solution:

Impedance, $Z = 19.7 + j21.1 = 28.85 \angle 47^\circ \Omega$

The instantaneous current,

$$i_s = v_s / Z = \frac{141.4}{28.85} \sin(\omega t - 47^\circ) = 4.9 \sin(\omega t - 47^\circ) \text{ A}$$

The expression for instantaneous power consumed by the circuit is,

$$\begin{aligned} p_s &= v_s i_s = 141.4 \sin \omega t \cdot 4.9 \sin(\omega t - 47^\circ) \\ &= 692.86 \sin \omega t [\sin \omega t \cos 47^\circ - \cos \omega t \sin 47^\circ] \\ &= 692.86 (\sin \omega t)^2 \cos 47^\circ - 692.86 (\sin \omega t \cos \omega t) \sin 47^\circ \\ &= 236.26 (1 - \cos 2\omega t) - 253.36 (\sin 2\omega t) \\ &= (236.26 - 236.26 \cos 2\omega t - 253.36 \sin 2\omega t) \text{ watts} \end{aligned}$$

In this expression,

$236.26 - 236.26 \cos 2\omega t$ = instantaneous real power ,

$- 253.36 \sin 2\omega t$ = instantaneous imaginary power

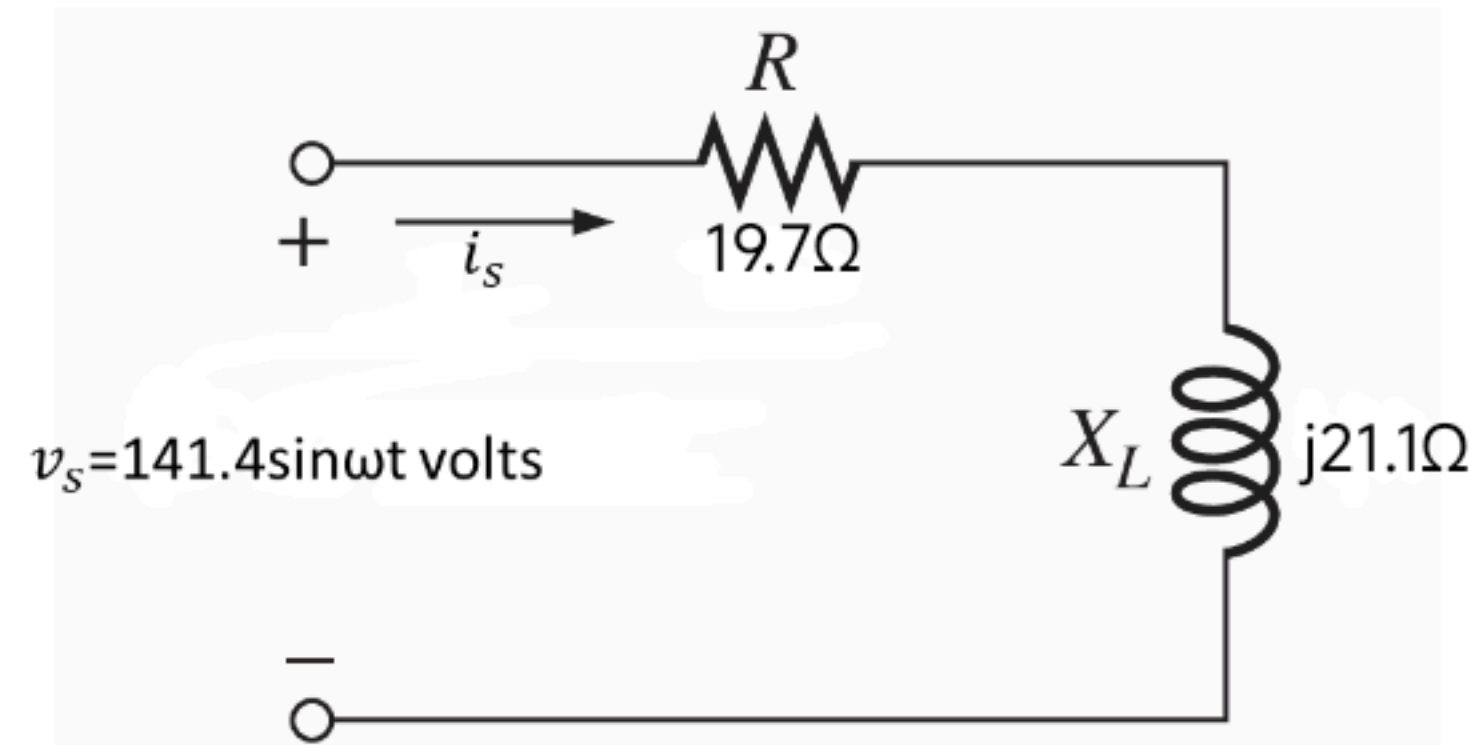
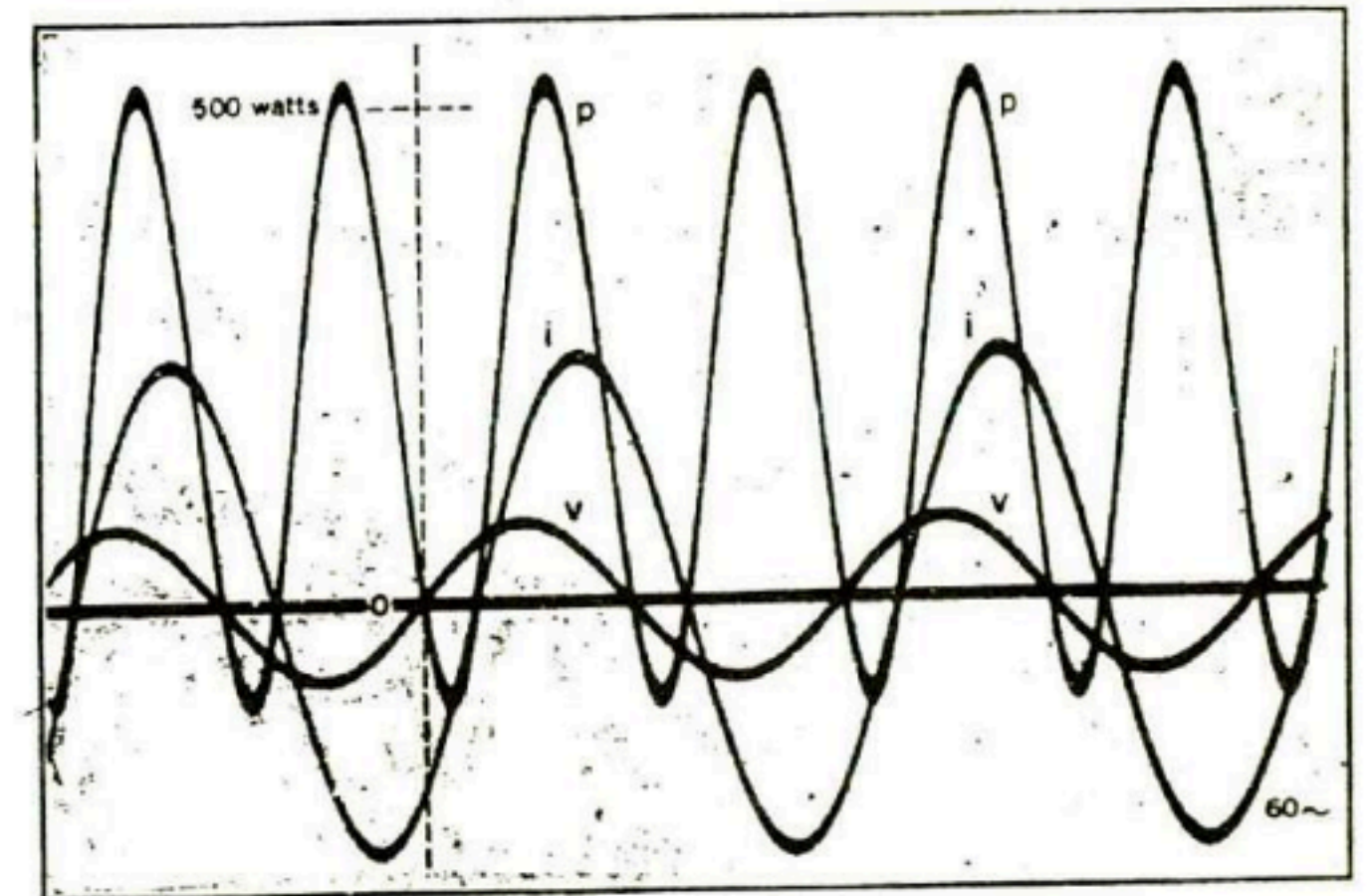


Fig.1



Assignment

2.30 A resistive element of 30 ohms is connected in series with an inductance coil, the self-inductance of which is 50 millihenrys and the ohmic resistance of which is 4.5 ohms. A voltage $v=100 \cos 377t$ volts is connected to the series branch.

- (a) Evaluate the expression for $i(t)$
- (b) Evaluate the expression for $p(t)$
- (c) Write the expression for the real power as a function of time, employing numerical coefficients. What is the average value of the instantaneous real power?
- (d) Write the expression for the reactive volt-amperes as a function of time, employing numerical coefficients. What is the average value of the instantaneous reactive power?

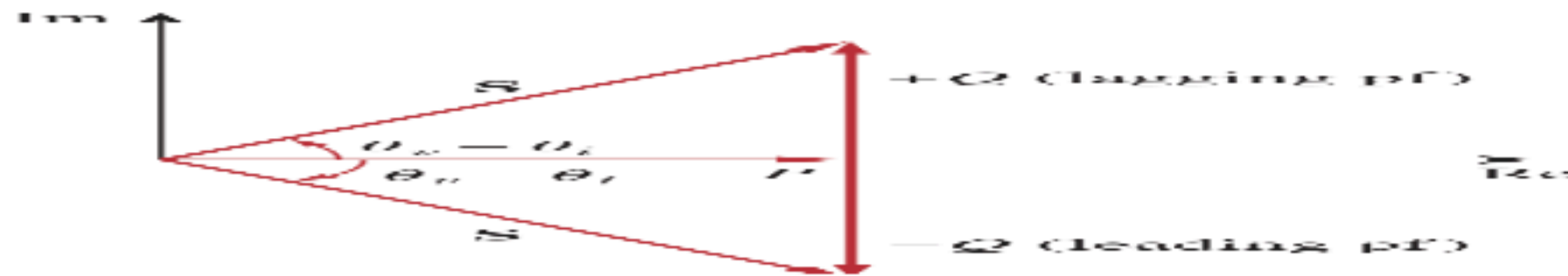


Complex power

Complex power (in VA) is the product of the rms voltage phasor and the complex conjugate of the rms current phasor. As a complex quantity, its real part is real power P and its imaginary part is reactive power Q .
Given the phasor form $V = V \angle \theta_v$ and $I = I \angle \theta_i$

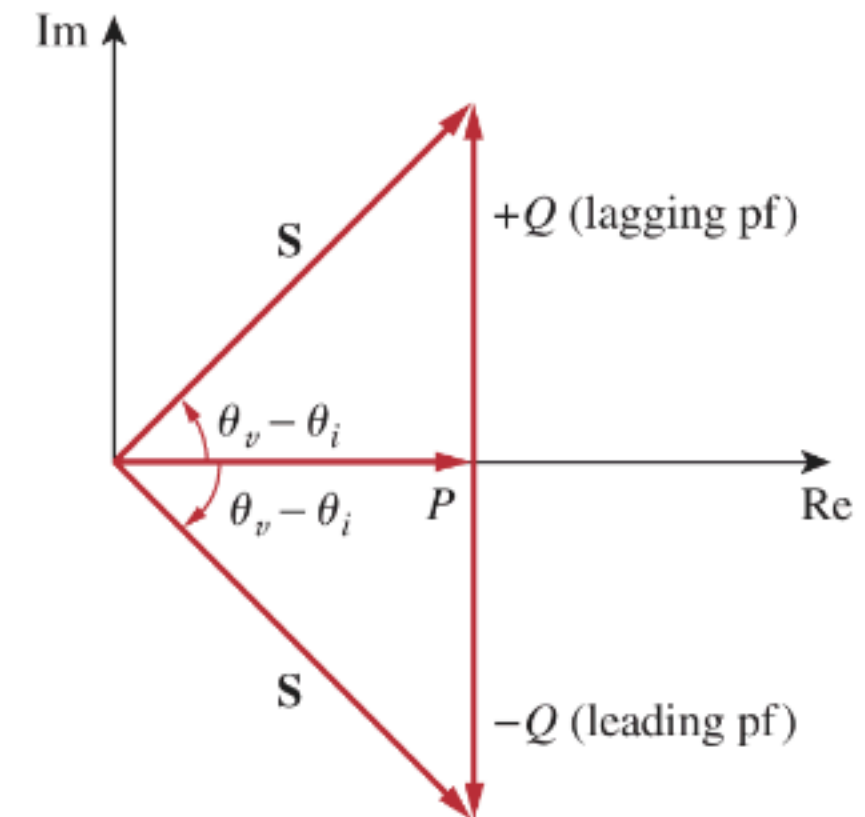
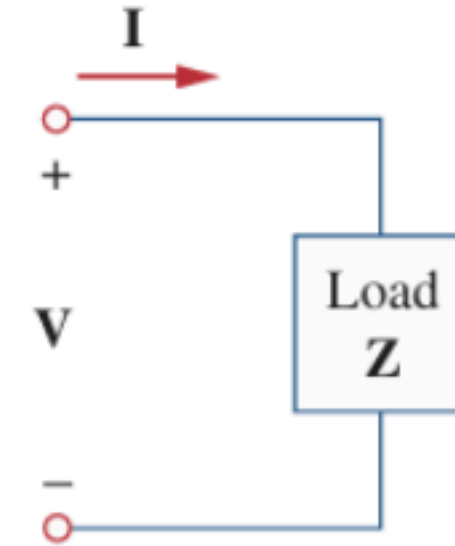
If V and I are Maximum value,
the complex power,

$$S = \frac{1}{2} VI^* = \frac{1}{2} (V \angle \theta_v)(I \angle -\theta_i) = \frac{1}{2} VI \cos(\theta_v - \theta_i) + j \frac{1}{2} VI \sin(\theta_v - \theta_i) = P + jQ$$



Notice that:

1. $Q = 0$ for resistive loads (unity pf).
2. $Q < 0$ for capacitive loads (leading pf).
3. $Q > 0$ for inductive loads (lagging pf).



Power triangle

Series Resonance

Series Resonance: A series circuit containing R, L, and C is in resonance when the resultant reactance is zero. Since the drop across the inductance leads the current by 90° whereas that across the condenser lags the current by 90° , the two drops are opposite. If they are made equal as in Fig.2, the reactive voltage drops neutralize and the impressed voltage is equal only to the resistance drop. This condition is called series resonance. Inspection of the vector diagram of Fig. 2 shows that the applied voltage is in phase with the current. The power factor is unity, and the circuit is in resonance. Thus for series resonance,

$$|X_L| = |X_C| \quad \text{or} \quad X_L = X_C$$

The series resonant frequency is

$$f_s = \frac{1}{2\pi\sqrt{LC}}$$

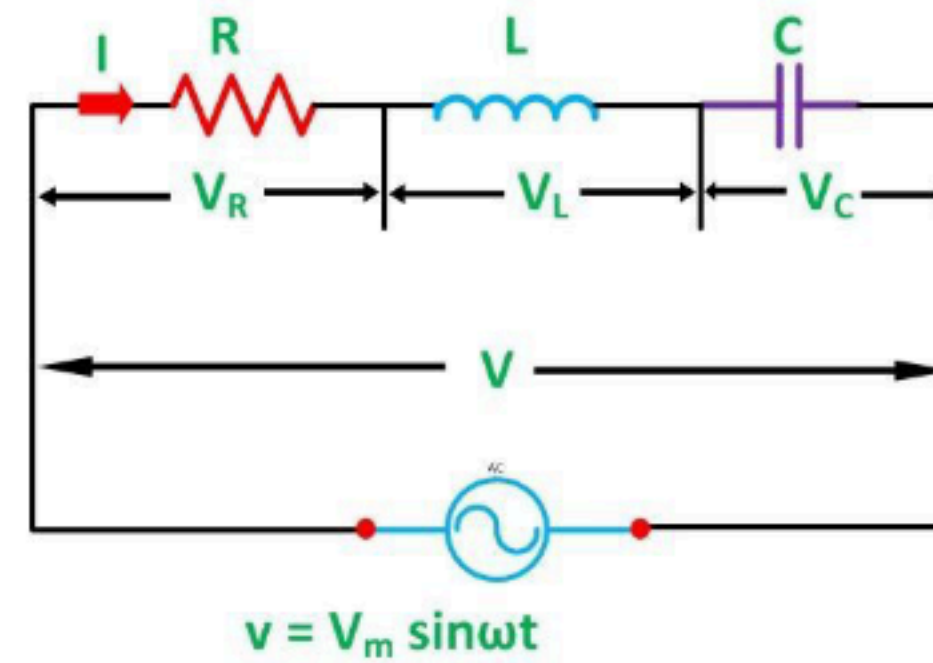


Fig.1

$$f_s = \frac{1}{2\pi\sqrt{LC}}$$

Fig.2

Series Resonance



The current is

$$I = \frac{V}{Z} = \frac{V}{\sqrt{R^2 + (X_L - X_C)^2}} = \frac{V}{\sqrt{R^2 + \left(2\pi fL - \frac{1}{2\pi fC}\right)^2}}$$

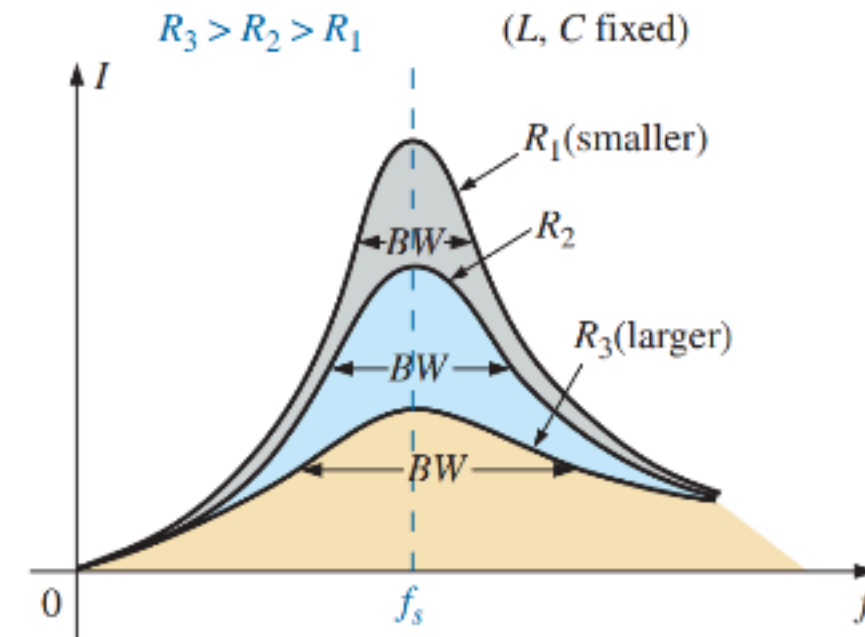
The voltage drop across the resistance, inductance and capacitance is

$$V_R = IR = \frac{VR}{\sqrt{R^2 + \left(2\pi fL - \frac{1}{2\pi fC}\right)^2}}$$

$$V_L = IX_L = \frac{VX_L}{\sqrt{R^2 + \left(2\pi fL - \frac{1}{2\pi fC}\right)^2}}$$

$$V_C = IX_C = \frac{VX_C}{\sqrt{R^2 + \left(2\pi fL - \frac{1}{2\pi fC}\right)^2}}$$

Effect of resistance on current variation in the range of series resonance:



In a series resonant circuit, the resistor controls how sharp or wide the response is. If you increase the resistance, the peak current gets smaller and the bandwidth becomes wider, making the curve broader. On the other hand, if you decrease the resistance, the peak current gets larger and the bandwidth becomes narrower, resulting in a sharper and more selective curve.

Series Resonance



The Effect of Varying Inductance (L) on Series Resonance:

When the inductance (L) in a series resonant circuit is varied, the voltage across the capacitor (V_C) and the inductor (V_L) behave differently. The maximum value of V_C occurs exactly at the resonant frequency because it depends directly on the current (I), which is also at its maximum at resonance. In contrast, the maximum voltage across the inductor (V_L) occurs at a frequency after (higher than) resonance. This is because V_L is the product of both the current (I) and the inductive reactance (X_L), which increases with frequency. Even after the peak current passes, the increasing X_L continues to push V_L higher until the point where the rate of decrease in current finally outweighs the rate of increase in reactance.

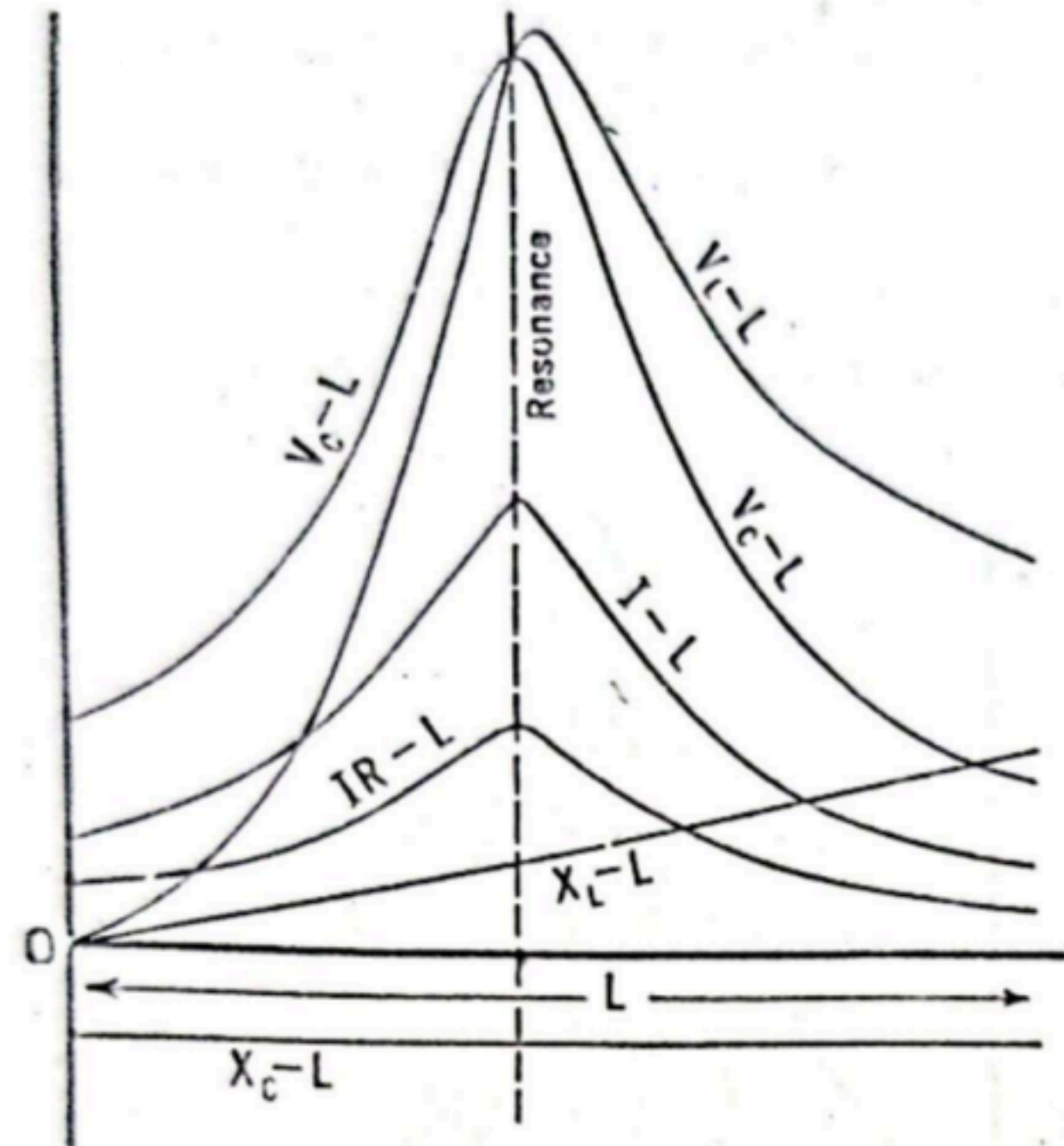


FIG. Series resonance by varying L .

Series Resonance



This specific point for maximum V_L can be found mathematically by setting the derivative dV_L/dX_L to zero.

$$\frac{dV_L}{dX_L} = \frac{[R^2 + (X_L - X_C)^2]^{1/2} V - V X_L \frac{1}{2} [R^2 + (X_L - X_C)^2]^{-1/2} 2(X_L - X_C)}{R^2 + (X_L - X_C)^2} = 0$$

and
$$X_L = \frac{R^2 + X_C^2}{X_C}$$

or
$$L = \frac{1}{2\pi f} \left(\frac{R^2 + X_C^2}{X_C} \right) = C(R^2 + X_C^2)$$

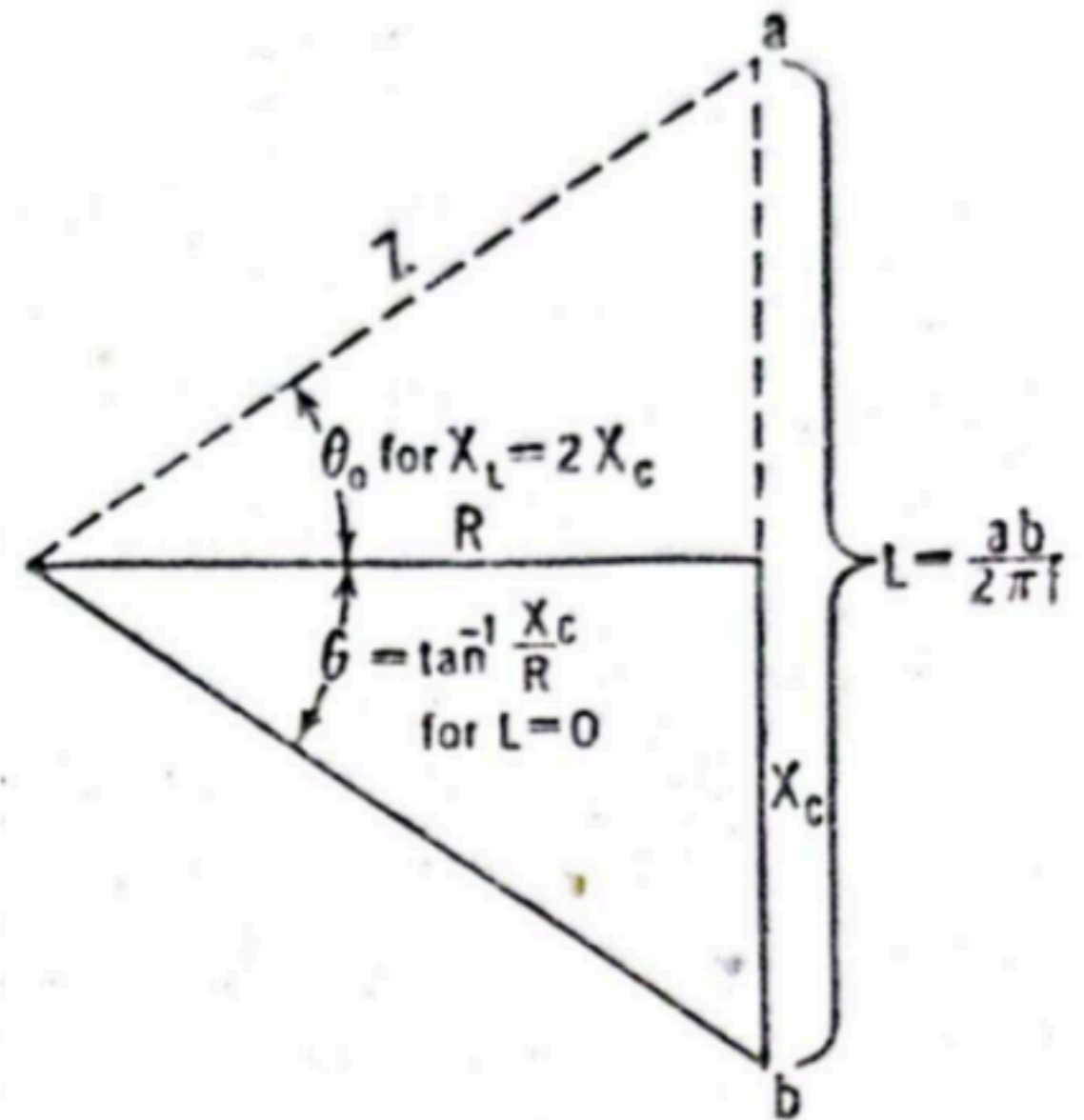


FIG. Impedance diagram showing the power factor angle θ as L is varied in an RLC series circuit.